T2K 1.431$\times 10^{20}$-POT Muon-Neutrino Disappearance Analysis

Costas Andreopoulos*[,a], Thomas Dealtry[a,b], James Dobson[d,a], Nick Grant[c], Martin Haigh[b], Jelena Ilic[a], Geoff Pearce[a], and Alfons Weber[a,b]

[a] *Particle Physics Dept., Rutherford Appleton Laboratory, STFC*
Harwell Science and Innovation Campus
Oxfordshire OX11 0QX, UK
[b] *Subdepartment of Particle Physics, University of Oxford*
Denys Wilkinson Building, Keble Road
Oxfordshire OX1 3RH, UK
[c] *Physics Dept., Lancaster University*
Lancaster, LA1 4YB, UK
[d] *Physics Dept., Imperial College London*
Blackett Lab, Prince Consort Road
London SW7 2BW, UK

Abstract

We report the results of a 2-flavour ν_μ-disappearance analysis on the combined Run1+2 dataset. This combined dataset corresponds to an integrated JPARC neutrino beam exposure of 1.431×10^{20} POT. The oscillation analysis was performed using both GENIE [1] and NEUT [2]. All of the inputs to the analysis are official 2010 inputs.

Using the ND280 measurement, GENIE predicts 101.72 $\pm$ 10.09 (stat) $^{+13.88}_{-13.49}$ (syst) 1 μ-like ring events in SuperK without oscillations but only 31 were observed. The statistical significance of the observed deficit is $\sim$4.2 standard deviations. NEUT predicts 103.59 $\pm$ 10.18 (stat) $^{+13.77}_{-13.43}$ (syst) events in SuperK, which implies that the statistical significance of the observed deficit is $\sim$4.3 standard deviations.

The observed number of 1μ-like ring events is consistent with the 2011 MINOS best-fit ν_μ-disappearance hypothesis ($sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32$\times 10^{-3}$ eV^2/c^4) [3]. Under this oscillation hypothesis, GENIE predicts 28.39 $\pm$ 5.33 (stat) $^{+4.53}_{-4.46}$ (syst) 1 μ-like ring events in SuperK, while NEUT predicts 29.18 $\pm$ 5.40 (stat) $^{+4.31}_{-4.26}$ (syst) events.

A 2-flavour ν_μ-disappearance fit to the observed reconstructed energy spectrum of 1 μ-like ring events of the combined Run1+2 dataset, with $sin^2 2\theta_{23}$ constrained to the physical region, yields $|\Delta m^2_{23}|$ = 2.60$\times 10^{-3}$ eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.97 as the best-fit values for GENIE (χ^2/ndf = 0.310). For NEUT, the corresponding values are $|\Delta m^2_{23}|$ = 2.65$\times 10^{-3}$ eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.98 (χ^2/ndf = 0.311).

*E-mail: costas.andreopoulos@stfc.ac.uk

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1. Introduction

In this technical note, we report the results of a 2-flavour ν_μ-disappearance analysis, performed by the Rutherford-Oxford-Lancs groups, on the combined Run1+2 (1.431×10^{20} POT) dataset. This dataset was collected in two run periods: in January 2010 - June 2010 (Run1, 0.323×10^{20} POT) and in November 2010 - March 2011 (Run2, 1.108×10^{20} POT).

This analysis [1] is an update (with larger statistics) of the work that was reported in Ref. [4], and reviewed by M.Nakahata et al. [5] in February 2011. This note is meant to be read in conjunction with Ref. [4] as the methodology is largely unchanged. The method for predicting the nominal SuperK reconstructed energy spectrum of 1 μ-like ring events is presented in Sections 2.1 - 2.6 of Ref. [4] and is unchanged. This analysis uses an updated SuperK flux tuning and an updated ND280 event-rate normalisation correction compared with those used to obtain the results in Ref. [4]. The flux normalisation (N_{sk}/N_{nd}) error, the flux shape error, the SuperK energy scale error and the SuperK ring-counting efficiency error have also been revised. All changes to the analysis inputs are summarised in Sec. 2 of this note. Updated nominal 1.431×10^{20} POT 1 μ-like ring event predictions are shown in Sec. 3 of this note.

As in the previous analysis, the oscillation fit was made using the likelihood ratio method [6], and the confidence regions were constructed using the Feldman-Cousins method [8]; these are unchanged and are described in Sections 3.1, 3.2 and 3.4 of Ref. [4]. Systematic errors were included in the oscillation contours using the Cousins-Highland method (Bayesian integration) and by determining their effects on the Feldman-Cousins $\Delta\chi^2_{critical}$ values as in Ref. [4]. The $\Delta\chi^2_{critical}$ values have been updated compared with Ref. [4] using the modified analysis inputs. This updated Feldman-Cousins construction is discussed in Sec. 5 of this note.

The results of the measurement of $\sin^2 2\theta_{23}$ and Δm^2_{23} using the Run1+2 dataset are given in Sec. 4 of this note for both GENIE and NEUT. The Feldman-Cousins confidence regions are shown in Sec. 6. The results from the fit to the combined Run1+2 dataset are compared with external measurements, the T2K sensitivity (for the MINOS 2011 best-fit oscillation parameters) and the results of separate fits to the Run1-only and Run2-only datasets.

Finally, a summary is given in section 7.

2. Summary of changes to analysis inputs compared with T2K-TN-036 (v4)

The following changes have been implemented compared with the ν_μ-disappearance analysis described in [4]:

- The SuperK flux has been upgraded from 10d(v2) to 10d(v3) [11]. When fitting the reconstructed energy spectrum of the combined Run1+2 single μ-like ring event sample, we used a POT-weighted Run1+Run2 flux in order to take into account the different conditions of Run1 and Run2.

- The flux normalisation (N_{sk}/N_{nd}) error has been updated. The updated error calculation code (v3.1) was obtained from [12]. Typical errors used in Ref. [4] were ∼9.7% without oscillations and ∼6.3% for oscillations with $sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32×10^{-3} eV^2/c^4. The updated flux normalisation errors used in this analysis for these two characteristic cases are ∼7.3% and ∼4.7% respectively.

- The flux shape error has been updated, and the updated flux fractional covariance matrix (v3.1) was obtained from [12].

- An updated ND280 event rate normalisation correction was used, and it was calculated using the 10d(v3) flux. The updated ND280 correction factor and energy spectra for various CC-like event categories were obtained from [14]. The ND280 normalisation correction was based on an analysis of the Run1 data only and, as before, it is energy independent.

- The ND280 efficiency error was increased from ±5% to +5.6% -5.2%. The statistical error of the observed number of CC-like events was also propagated to SuperK.

- The SuperK ring-counting (RC) efficiency errors have been revised [15]. The CCQE RC error has been revised from 7.6% (no energy dependence) to 1.7% for $E_{reco} < 0.4$ GeV, 3.5% for 0.4 GeV $\leq E_{reco} \leq$ 1.1GeV and 9.3% for $E_{reco} > 1.1$ GeV. The bin-to-bin correlations in the RC efficiency error in the three reconstructed neutrino energy bins have also been taken into account, and the covariance matrix is given in Tab. 1. For CCnonQE the RC efficiency error has been revised from 25.4% to 19.9% (no energy dependence) and for NC the RC efficiency error has been revised from 57.0% to 48.2% (no energy dependence). All other SuperK efficiency errors were unchanged.

- The SuperK energy scale error has been revised from 1% to 2.3% [16].

[1] Preliminary results from this work were presented on May 20th 2011 at the T2K collaboration meeting at KEK [10]. Sensitivity studies, performed before looking at the Run1+2 dataset, were presented on March 31st 2011 [9].

A summary of all the sources of systematic error considered in this analysis is given in Tab. 1; this is an update of Tab. 6 in Ref. [4]. The treatment of these uncertainties is largely unchanged and is described in detail in Sec. 4.1 and Eq. 31 of Ref. [4]. The only technical complication is that the SuperK CCQE efficiency error (which was 7.8% independent of energy in [4]) has been split into two parts: these are the RC efficiency error and the 'other' efficiency error (i.e. fiducial volume, reduction, OD cut, PID, momentum cut and decay electron cut). The CCQE RC efficiency error now has a dependence on the reconstructed neutrino energy and the bin-to-bin correlations are taken into account. The oscillation analysis code has been updated to include these changes. The bin-to-bin correlations in the RC efficiency error are propagated using the Cholesky method [7]; this method was employed in the earlier analysis to propagate the correlated systematic errors in the flux shape (see Sec. 4.5 of Ref. [4]).

3. Updated predictions of numbers of single μ-like ring events in SuperK

In this section we present plots and tables comparing the GENIE and NEUT predictions (after extrapolation) of the numbers of 1 μ-like ring events in SuperK under various oscillation hypotheses.

In Figs. 1 and 2 are shown, respectively, the GENIE and NEUT predicted (after extrapolation) non-oscillated reconstructed-energy spectrum of 1 μ-like ring events passing the so-called 'tight' cuts (see Tab. 1 of Ref. [4]). The contributions from various true neutrino reaction modes are also shown. Figs. 3 and 4 show the same distribution with a log scale so that the contribution from all non-ν_μ CCQE backgrounds can be seen. The corresponding predictions for oscillations with typical parameters ($sin^2 2\theta_{23}$=1.0 and $|\Delta m^2_{23}|$=2.32$\times 10^{-3}$ eV^2/c^4) are shown in Figs. 5 and 7 for GENIE and Figs. 6 and 8 for NEUT.

A summary is given in Tab. 2 for both oscillated and non-oscillated scenarios of the expected numbers of events (after extrapolation) in SuperK and the nominal expected contributions from various true neutrino reaction modes (corresponding to the different SuperK MC templates used in this analysis, see Sec. 2.4 of Ref. [4]).

In Fig. 9 we show the GENIE prediction of the integrated number of 1 μ-like ring events (after extrapolation) for an extended grid of $sin^2 2\theta_{23}$ and Δm^2_{23} values. The corresponding plot for NEUT is shown in Fig. 10. The same plots are shown in Fig. 11 (GENIE) and Fig. 12 (NEUT) but zoomed in around the MINOS 2011 best-fit values.

The estimated uncertainty in the number of 1 μ-like ring SuperK events (after extrapolation) due to all the sources of uncertainty listed in Tab. 1 is shown in for both GENIE and NEUT in Tab. 3 (no oscillation) and Tab. 4 (oscillations with $sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32$\times 10^{-3}$ eV^2/c^4). The error envelope of the 1 μ-like ring event reconstructed-energy spectrum (after extrapolation) for the latter typical oscillation hypothesis is shown in Fig. 13 for GENIE and in Fig. 14 for NEUT.

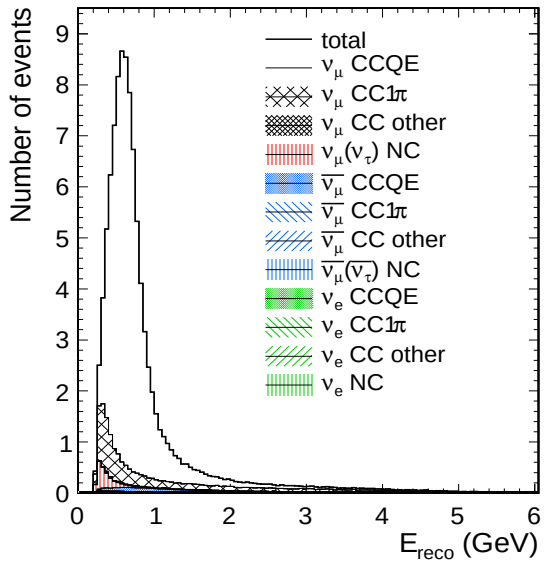


Figure 1: Predicted non-oscillated reconstructed-energy spectrum of 1 μ-like ring events (after extrapolation) and contributions from various true neutrino reaction modes (GENIE)

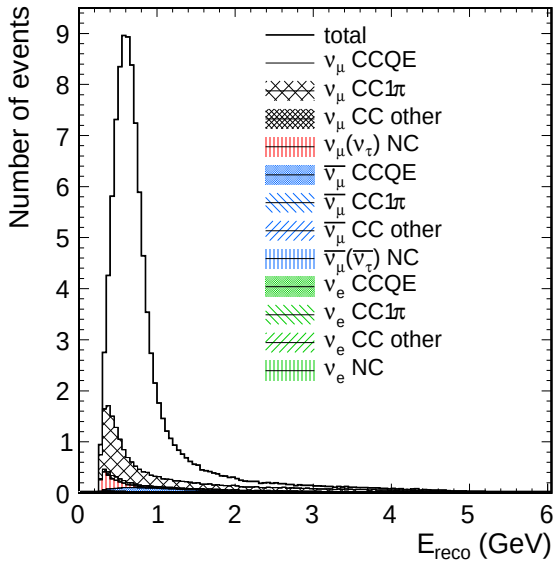


Figure 2: Predicted non-oscillated reconstructed-energy spectrum of 1 μ-like ring events (after extrapolation) and contributions from various true neutrino reaction modes (NEUT)

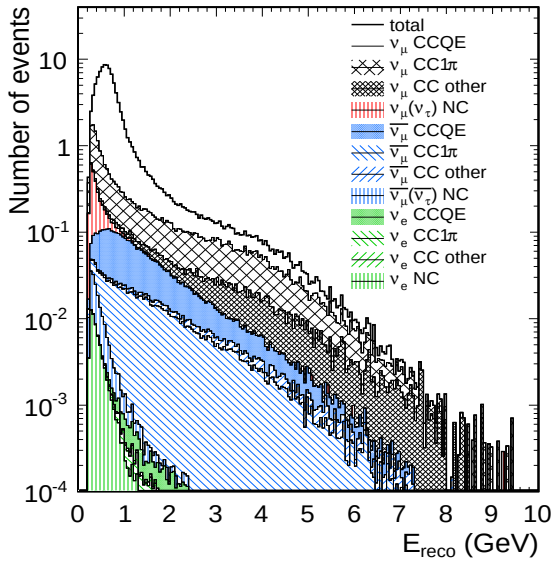


Figure 3: Predicted non-oscillated reconstructed-energy spectrum of 1 μ-like ring events (after extrapolation) and contributions from various true neutrino reaction modes (GENIE)

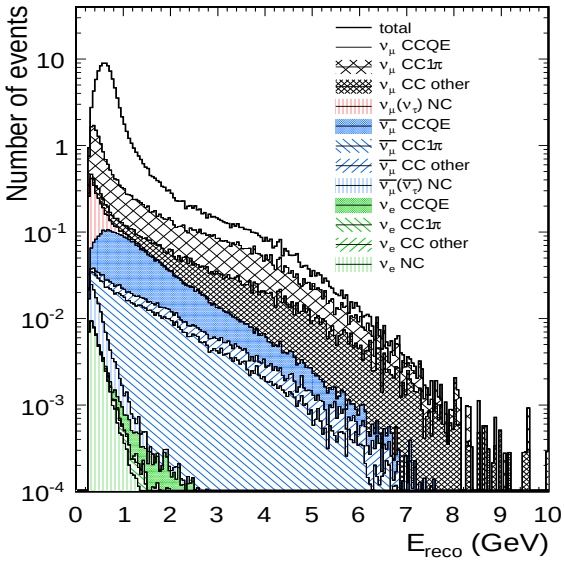


Figure 4: Predicted non-oscillated reconstructed-energy spectrum of 1 μ-like ring events (after extrapolation) and contributions from various true neutrino reaction modes (NEUT)

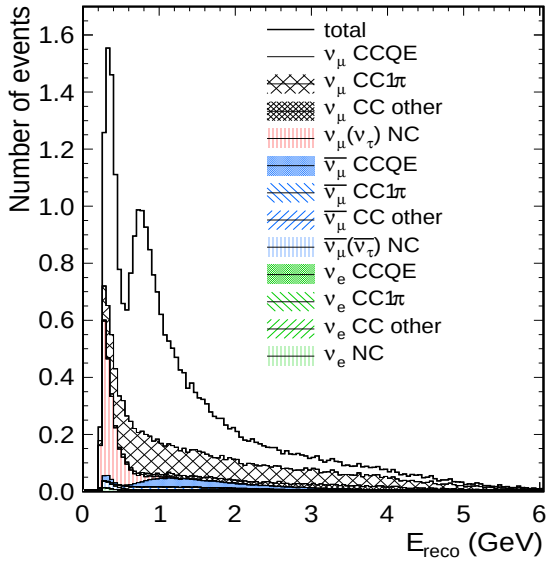


Figure 5: Predicted oscillated ($sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32×10^{-3} eV^2/c^4) reconstructed-energy spectrum of 1 μ-like ring events (after extrapolation) and contributions from various true neutrino reaction modes (GENIE)

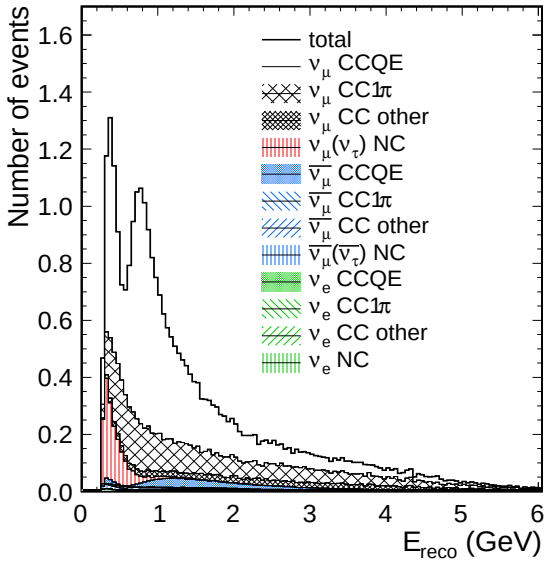


Figure 6: Predicted oscillated ($sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32×10^{-3} eV^2/c^4) reconstructed-energy spectrum of 1 μ-like ring events (after extrapolation) and contributions from various true neutrino reaction modes (NEUT)

SuperK detector uncertainties	
$f^{SK}_{\epsilon;r}$	Combined uncertainty in efficiency due to uncertainties in fiducial volume, reduction chain, OD cut, RC cut, PID cut, momentum cut and decay electron cut [15]. Applied to corresponding SuperK MC templates: • CCQE: – RC: 1.7% ($E_{reco} <$ 0.4 GeV), 3.5% (0.4 GeV $\leq E_{reco} \leq$ 1.1 GeV), 9.3% ($E_{reco} >$ 1.1 GeV). Error covariance matrix is $\begin{bmatrix} (1.7\%)^2 & 0.59*1.7\%*3.5\% & 0.33*1.7\%*9.3\% \\ & (3.5\%)^2 & 0.61*3.5\%*9.3\% \\ & & (9.3\%)^2 \end{bmatrix}$ – other: 1.7% (energy-independent) • CCnonQE: 19.9% RC + 1.7% other ∼20% (energy-independent) • NC: 48.2% RC + 100% PID + 1.7% other ∼111% (energy-independent) • ν_e CC: ∼100% (energy-independent)
f^{SK}_{E}	Reconstructed energy scale error is 2.3% [16]. Applied to SuperK MC templates by scaling bin edges and, assuming uniform distribution of events within the 50 MeV bins, calculating the number of events gained from / lost to neighbouring bins.
ND280 detector uncertainties	
f^{ND280}_{ϵ}	Efficiency error is +5.6% -5.2% [14]. Applied to ND280 MC templates.
Flux uncertainties	
f_{norm}	Flux normalization uncertainty. This uncertainty, expressed relative to the N_{SK}/N_{ND280} ratio predicted by the MC, was calculated for oscillation and other systematic parameters using code provided by the Beam WG [12]. The value obtained for nominal MC and oscillations with parameters $sin^2 2\theta_{23}$ = 1.00 and $\lvert\Delta m^2_{23}\rvert$ = 2.32×10^{-3} eV^2/c^4 was ∼4.7% (∼7.3% without oscillations).
$\rho^{\mathbf{\Phi_{SK}}}$	Flux shape uncertainty. Expressed in terms of true energy bin-to-bin correlation matrix for SuperK ν_μ flux provided by the Beam WG [13]. Uncertainty was propagated using the Choleskly method [7].
Neutrino Generator (GENIE/NEUT) model uncertainties	
$f_{\sigma_{CCQE};t}$	CCQE cross section uncertainty. This respresents only the "residual" CCQE cross section uncertainty that is not cancelled in the near-to-far extrapolation; this residual uncertainty is due to the different flux energy ranges sampled by the near and far detectors and the different compositions of their nuclear targets. Therefore the uncertainty is applied to the SuperK MC templates only. This uncertainty has dependence on true neutrino energy [17].
$f_{\sigma_{CC1\pi}/\sigma_{CCQE};t}$	Uncertainty in CC1π/CCQE cross section ratio: 30% for true energy $<$ 2 GeV and 20% for true energy $>$ 2 GeV [17]. Applied to both corresponding SuperK and ND280 MC templates.
$f_{\sigma_{CCoth}/\sigma_{CCQE};t}$	Uncertainty in CCother/CCQE cross section ratio: 30% for true energy $<$ 2 GeV and 25% for true energy $>$ 2 GeV [17]. Applied to both corresponding SuperK and ND280 MC templates.
$f_{\sigma_{NC}/\sigma_{CCQE}}$	Uncertainty in NC/CCQE cross section ratio: 36% [17]. Applied to both corresponding SuperK and ND280 MC templates.
$f_{FSI;r}$	Uncertainty in effect of FSI (final state interaction / intranuclear hadron transport) to exclusive CCQE-like (1 μ-like ring) SuperK measurement (e.g. effect of pion absorption in oxygen). Applied to SuperK MC templates. The error has dependence on reconstructed energy [17].

Table 1: Inputs for ν_μ-disappearance systematic error evaluation (this is an updated version of Tab. 6 shown in Ref. [4]).

	(a)		(b)	
	GENIE	NEUT	GENIE	NEUT
Total	101.717	103.593	28.394	29.177
ν_μ CCQE	82.074 (80.688%)	82.348 (79.491%)	16.520 (58.182%)	16.731 (57.343%)
ν_μ CC1π	12.895 (12.677%)	13.986 (13.501%)	6.676 (23.510%)	7.014 (24.040%)
ν_μ CCother	1.315 (1.292%)	2.306 (2.226%)	1.107 (3.900%)	1.800 (6.168%)
$\nu_\mu(\nu_\tau)$ NC	2.116 (2.080%)	1.682 (1.624%)	2.116 (7.452%)	1.682 (5.766%)
$\bar{\nu_\mu}$ CCQE	2.111 (2.076%)	2.186 (2.110%)	1.045 (3.679%)	1.112 (3.813%)
$\bar{\nu_\mu}$ CC1π	0.885 (0.870%)	0.698 (0.674%)	0.634 (2.231%)	0.510 (1.748%)
$\bar{\nu_\mu}$ CCother	0.131 (0.129%)	0.227 (0.219%)	0.106 (0.371%)	0.167 (0.572%)
$\bar{\nu_\mu}(\bar{\nu_\tau})$ NC	0.117 (0.114%)	0.093 (0.090%)	0.117 (0.411%)	0.093 (0.319%)
ν_e CCQE	0.007 (0.007%)	0.008 (0.007%)	0.007 (0.025%)	0.008 (0.026%)
ν_e CC1π	0.004 (0.004%)	0.005 (0.005%)	0.004 (0.015%)	0.005 (0.016%)
ν_e CCother	0.001 (0.001%)	0.002 (0.002%)	0.001 (0.004%)	0.002 (0.006%)
ν_e NC	0.062 (0.061%)	0.054 (0.052%)	0.062 (0.220%)	0.054 (0.184%)

Table 2: Expected numbers of SuperK 1 μ-like ring events from GENIE and NEUT (after extrapolation) for (a) no oscillation and (b) oscillations with $sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}| = 2.32\times10^{-3}\ eV^2/c^4$.

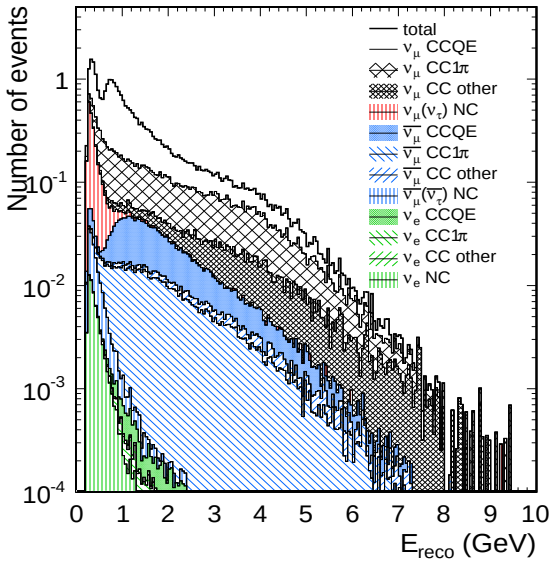


Figure 7: Predicted oscillated ($sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}| = 2.32\times10^{-3}$ eV^2/c^4) reconstructed-energy spectrum of 1 μ-like ring events (after extrapolation) and contributions from various true neutrino reaction modes (GENIE)

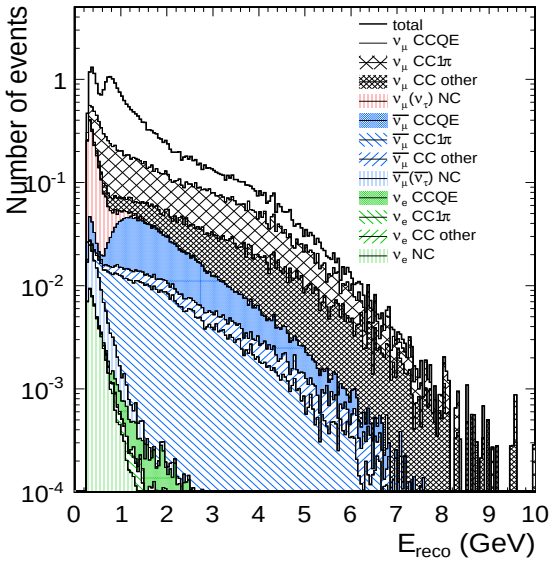


Figure 8: Predicted oscillated ($sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}| = 2.32\times10^{-3}$ eV^2/c^4) reconstructed-energy spectrum of 1 μ-like ring events (after extrapolation) and contributions from various true neutrino reaction modes (NEUT)

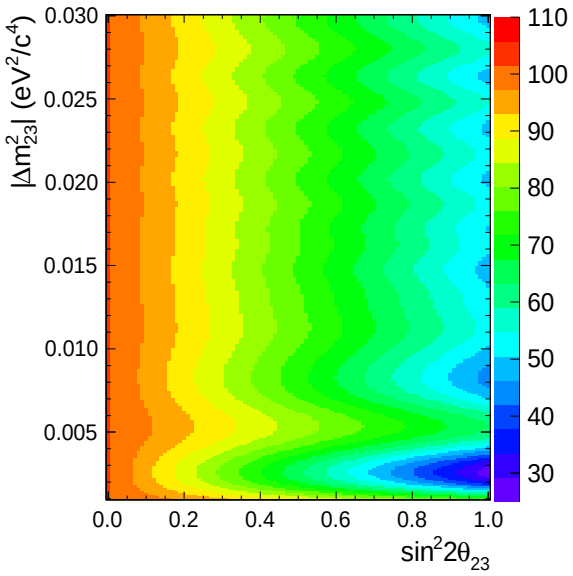


Figure 9: Expected number of 1 μ-like ring events (after extrapolation) in the 0-10 GeV reconstructed energy range (GENIE)

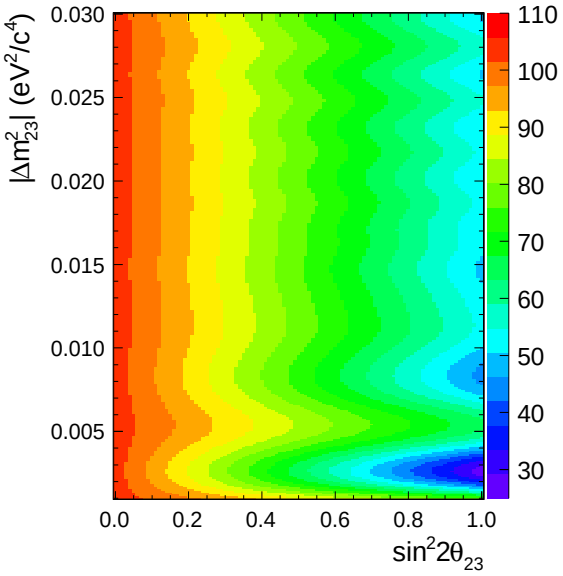


Figure 10: Expected number of 1 μ-like ring events (after extrapolation) in the 0-10 GeV reconstructed energy range (NEUT)

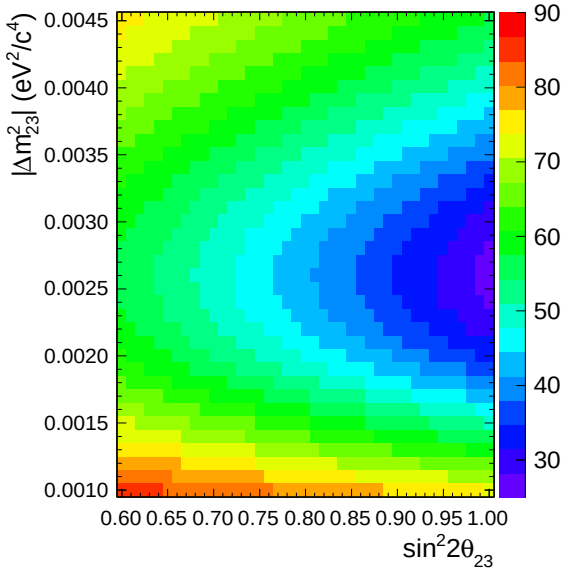


Figure 11: Expected number of 1 μ-like ring events (after extrapolation) in the 0-10 GeV reconstructed energy range, zoomed around the MINOS 2011 best-fit values (GENIE)

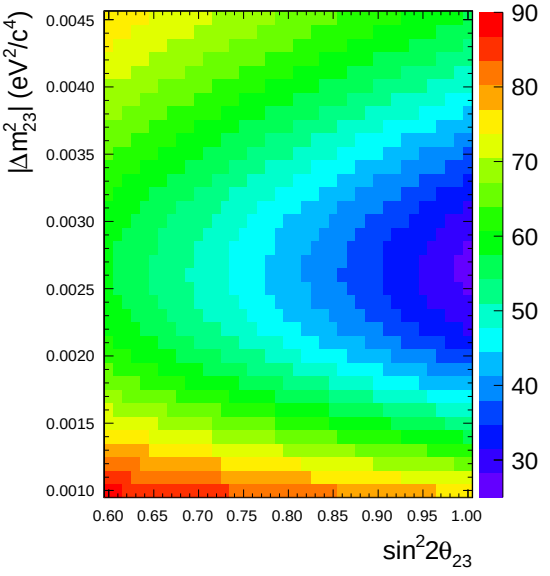


Figure 12: Expected number of 1 μ-like ring events (after extrapolation) in the 0-10 GeV reconstructed energy range, zoomed around the MINOS 2011 best-fit values (NEUT)

Error source	GENIE $\delta N_{SK}^{exp}/N_{SK}^{exp}$	NEUT $\delta N_{SK}^{exp}/N_{SK}^{exp}$
SuperK CCQE efficiency (ring-counting)	±2.99%	±3.06%
SuperK CCQE efficiency (other)	±1.41%	±1.39%
SuperK CCnonQE efficiency	±2.99%	±3.32%
SuperK NC efficiency	±2.51%	±1.96%
SuperK ν_e CC efficiency	±0.01%	±0.01%
ND280 efficiency	+5.49% -5.30%	+5.49% -5.30%
ND280 normalization	±2.63%	±2.62%
Flux normalization	±7.29%	±7.33%
CCQE cross section	±4.28%	±4.05%
CC1pi/CCQE cross section ratio	+2.66% -2.35%	+2.18% -1.93%
CCother/CCQE cross section ratio	+5.80% -5.16%	+5.27% -4.71%
NC/CCQE cross section ratio	+0.68% -0.66%	+0.81% -0.78%
FSI	±3.23%	±3.22%
Total	+13.65% -13.26%	+13.30% -12.97%

Table 3: Effect of 1σ systematic parameter variation on the non-oscillated number of 1 μ-like ring events predicted by GENIE and NEUT (after extrapolation)

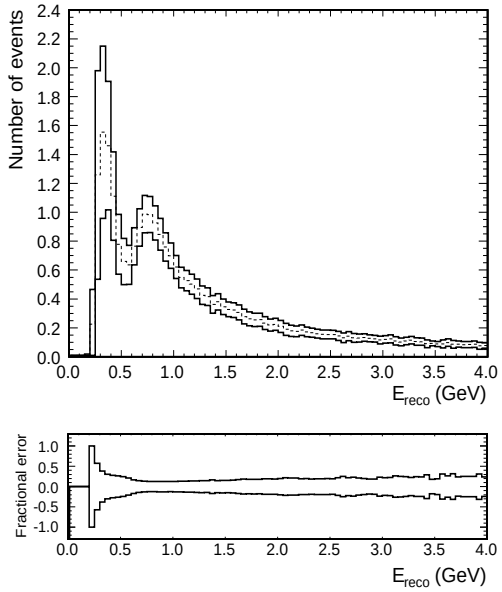


Figure 13: Error envelope of 1 μ-like ring event reconstructed-energy spectrum (after extrapolation) for oscillations with $sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32×10^{-3} eV^2/c^4. Calculated from toy MC experiments as the $\pm 1\sigma$ spread of bin contents (GENIE).

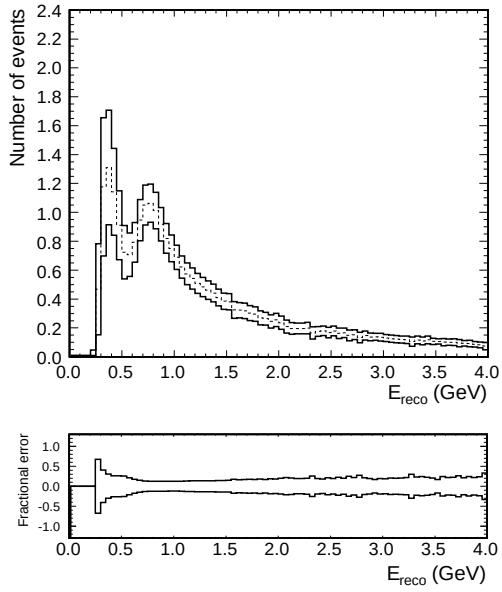


Figure 14: Error envelope of 1 μ-like ring event reconstructed-energy spectrum (after extrapolation) for oscillations with $sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32×10^{-3} eV^2/c^4. Calculated from toy MC experiments as the $\pm 1\sigma$ spread of bin contents (NEUT).

Error source	GENIE $\delta N_{SK}^{exp}/N_{SK}^{exp}$	NEUT $\delta N_{SK}^{exp}/N_{SK}^{exp}$
SuperK CCQE efficiency (ring-counting)	±3.04%	±3.23%
SuperK CCQE efficiency (other)	±1.05%	±1.04%
SuperK CCnonQE efficiency	±6.00%	±6.51%
SuperK NC efficiency	±8.97%	±6.96%
SuperK ν_e CC efficiency	±0.04%	±0.05%
ND280 efficiency	+5.49% -5.30%	+5.59% -5.30%
ND280 normalization	±2.63%	±2.62%
Flux normalization	±4.78%	±4.73%
CCQE cross section	±2.78%	±2.36%
CC1pi/CCQE cross section ratio	+0.22% -0.25%	+0.52% -0.59%
CCother/CCQE cross section ratio	+5.04% -4.48%	+4.10% -3.67%
NC/CCQE cross section ratio	+1.40% -1.44%	+0.82% -0.84%
FSI	±5.68%	±5.80%
Total	+15.94% -15.72%	+14.79% -14.61%

Table 4: Effect of 1σ systematic parameter variation on the number of 1 μ-like ring events predicted by GENIE and NEUT (after extarpolation) assuming oscillations with typical (MINOS 2011) parameters ($sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}| = 2.32\times10^{-3}\ eV^2/c^4$).

4. 2-flavour ν_μ-disappearance fits to the Run1, Run2 and Run1+2 datasets

Using the method outlined in the previous sections and in Ref. [4], a 2-flavour ν_μ-disappearance fit was performed on the 1.431×10^{20} POT Run1+2 dataset. For GENIE, this analysis found $|\Delta m^2_{23}|$ = 2.60$\times 10^{-3}$ eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.97 to be the best-fit values (χ^2/ndf=0.310). For NEUT, the corresponding best-fit values were $|\Delta m^2_{23}|$ = 2.65$\times 10^{-3}$ eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.98 (χ^2/ndf=0.311). The reconstructed neutrino energy distribution of the 31 single μ-like ring events in the Run1+2 dataset, along with the best-fit predictions from GENIE and NEUT, are shown in Figs. 15 and 16. These distributions are shown in the 50 MeV reconstructed neutrino energy bins that were used in the fitting procedure. It is illuminating to see the same distributions with a smaller number of bins of variable size in reconstructed neutrino energy; these are shown in Figs. 17 and 18, though this binning scheme was not used in the analysis. In Figs. 19 and 20, the ν_μ-CCQE component of the best-fit oscillation hypothesis is added to the distributions shown in the previous figures.

The $\Delta\chi^2$ distributions for the fit to the Run1+2 dataset are shown as a function of $sin^2 2\theta_{23}$ and $|\Delta m^2_{23}|$ in Fig. 21 for GENIE and Fig. 22 for NEUT. Meanwhile zoomed views around the Run1+2 best-fit values of these $\Delta\chi^2$ distributions are shown in Fig. 23 for GENIE and Fig. 24 for NEUT. The marginalised $\Delta\chi^2$ distributions for the Run1+2 fit are shown in Fig. 25 for GENIE and Fig. 26 for NEUT. The 1-dimensional marginalised $\Delta\chi^2$ distributions are obtained by minimizing $\Delta\chi^2$ with respect to $sin^2 2\theta_{23}$ for each $|\Delta m^2_{23}|$ bin (and vice versa).

Comparisons of the Run1+2 best-fit values with distributions obtained by 2-flavour ν_μ-disappearance fits on 1.431$\times 10^{20}$ POT toy MC experiments generated with $sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32$\times 10^{-3}$ eV^2/c^4 are shown in Figs. 27 and 29 for GENIE and in Figs. 28 and 30 for NEUT.

For comparison, we performed the same 2-flavour ν_μ-disappearance fit separately on the 0.323×10^{20}-POT Run1 and 1.108×10^{20}-POT Run2 datasets. A summary of the best-fit values for all fits is given in Table 5. The reconstructed neutrino energy spectrum for Run1 (Run2) 1 μ-like ring events along with the best-fit and null hypothesis predictions from GENIE are shown in Fig. 31 (Fig. 33). The corresponding Run1 (Run2) distributions for NEUT are shown in Fig. 32 (Fig. 34).

The marginalised $\Delta\chi^2$ distributions for the 2-flavour ν_μ-disappearance fits to the Run1, Run2 and the combined Run1+2 datasets are shown as a function of $sin^2 2\theta_{23}$ in Fig. 39 for GENIE and Fig. 40 for NEUT. These distributions are also shown as a function of $|\Delta m^2_{23}|$ in Fig. 35 for GENIE and Fig. 36 for NEUT (with $|\Delta m^2_{23}|$ restricted in the fit to the range 1$\times 10^{-3}$ eV^2/c^4 - 6$\times 10^{-3}$ eV^2/c^4). The effects on these distributions of extending the $|\Delta m^2_{23}|$ range to 1$\times 10^{-3}$ eV^2/c^4 - 30$\times 10^{-3}$ eV^2/c^4 are shown in Fig. 37 for GENIE and Fig. 38 for NEUT.

Dataset	GENIE best-fit values	NEUT best-fit values
0.323×10^{20} POT Run1	$\|\Delta m^2_{23}\|$ = 2.77 $\times 10^{-3}$ eV^2/c^4 $sin^2 2\theta_{23}$ = 0.81 χ^2/ndf = 0.105 (MINOS 2011 params χ^2/ndf = 0.121)	$\|\Delta m^2_{23}\|$ = 2.88 $\times 10^{-3}$ eV^2/c^4 $sin^2 2\theta_{23}$ = 0.84 χ^2/ndf = 0.114 (MINOS 2011 params χ^2/ndf = 0.129)
1.108×10^{20} POT Run2	$\|\Delta m^2_{23}\|$ = 2.59 $\times 10^{-3}$ eV^2/c^4 $sin^2 2\theta_{23}$ = 1.00 χ^2/ndf = 0.318 (MINOS 2011 params χ^2/ndf = 0.327)	$\|\Delta m^2_{23}\|$ = 2.62 $\times 10^{-3}$ eV^2/c^4 $sin^2 2\theta_{23}$ = 1.00 χ^2/ndf = 0.310 (MINOS 2011 params χ^2/ndf = 0.322)
1.431×10^{20} POT Run1+2	$\|\Delta m^2_{23}\|$ = 2.60 $\times 10^{-3}$ eV^2/c^4 $sin^2 2\theta_{23}$ = 0.97 χ^2/ndf = 0.310 (MINOS 2011 params χ^2/ndf = 0.319)	$\|\Delta m^2_{23}\|$ = 2.65 $\times 10^{-3}$ eV^2/c^4 $sin^2 2\theta_{23}$ = 0.98 χ^2/ndf = 0.311 (MINOS 2011 params χ^2/ndf = 0.322)

Table 5: Summary of best-fit parameter values for 2-flavour oscillation fits to the Run1, Run2 and the combined Run1+2 datasets. The χ^2/ndf (ndf=198) value is also given assuming the MINOS 2011 best-fit parameters ($sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32$\times 10^{-3}$ eV^2/c^4).

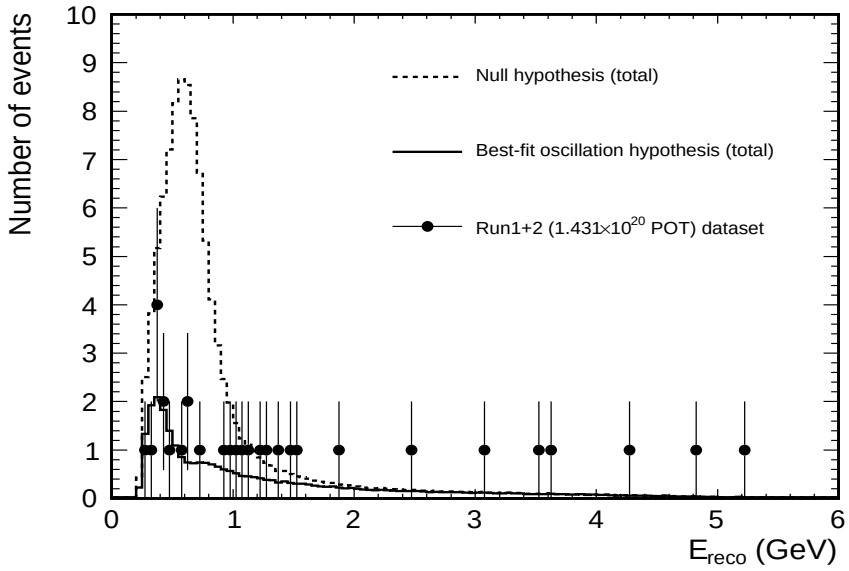


Figure 15: Reconstructed neutrino energy distribution of the 31 single μ-like ring events of the combined 1.431×10^{20} POT Run1+2 dataset (points) and GENIE best-fit SuperK prediction (solid line). The distributions are shown using the 50 MeV reconstructed neutrino energy binning used in the fitting procedure. The best-fit oscillation parameters are $|\Delta m^2_{23}|$ = 2.60×10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.97. The dotted line shows the GENIE prediction without oscillations.

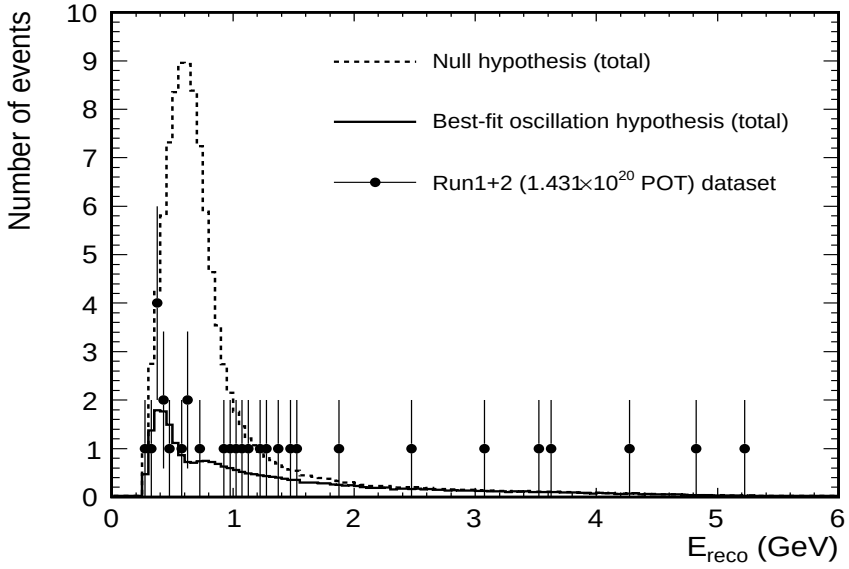


Figure 16: Reconstructed neutrino energy distribution of the 31 single μ-like ring events of the combined 1.431×10^{20} POT Run1+2 dataset (points) and NEUT best-fit SuperK prediction (solid line). The distributions are shown using the 50 MeV reconstructed neutrino energy binning used in the fitting procedure. The best-fit oscillation parameters are $|\Delta m^2_{23}|$ = 2.65×10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.98. The dotted line shows the NEUT prediction without oscillations.

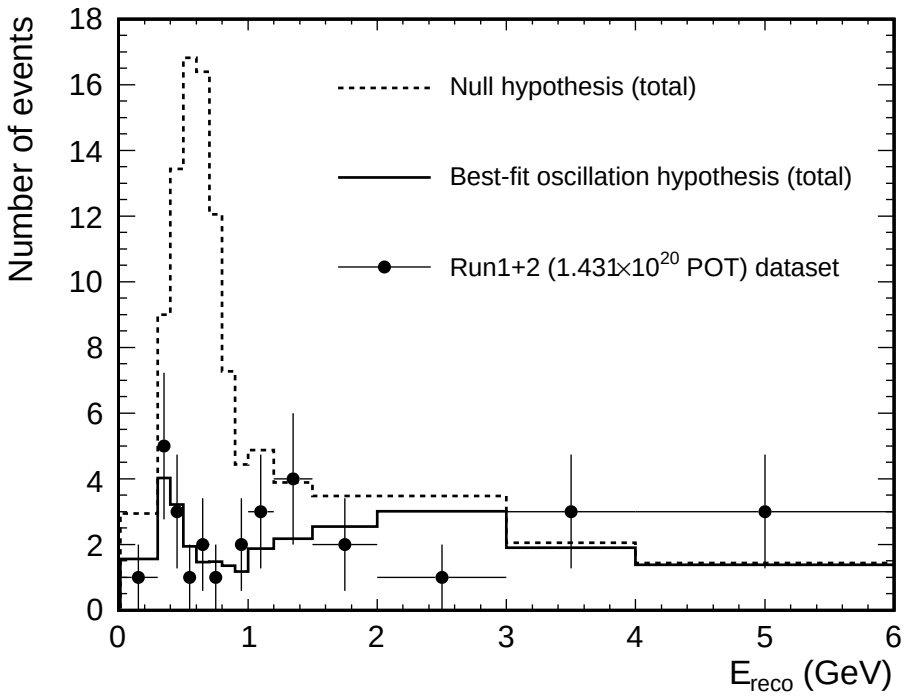


Figure 17: Reconstructed neutrino energy distribution of the 31 single μ-like ring events of the combined 1.431×10^{20} POT Run1+2 dataset (points) and GENIE best-fit SuperK prediction (solid line). The distributions are shown using a variable binning scheme; this is for illustration purposes only, and this binning scheme was not used in the analysis. The best-fit oscillation parameters are $|\Delta m^2_{23}|$ = 2.60×10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.97. The dotted line shows the GENIE prediction without oscillations.

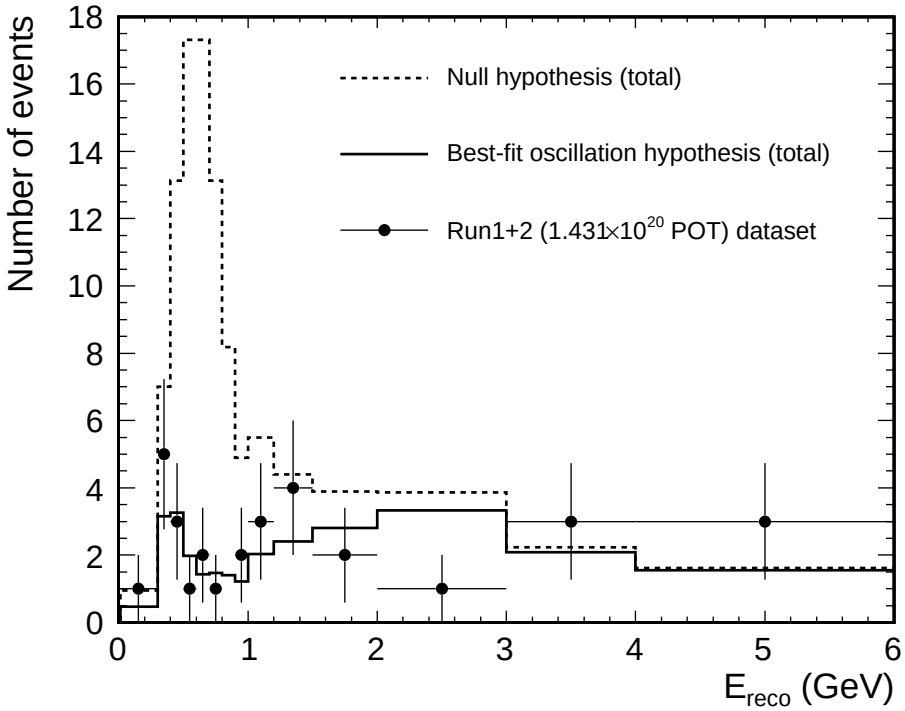


Figure 18: Reconstructed neutrino energy distribution of the 31 single μ-like ring events of the combined 1.431×10^{20} POT Run1+2 dataset (points) and NEUT best-fit SuperK prediction (solid line). The distributions are shown using a variable binning scheme; this is for illustration purposes only, and this binning scheme was not used in the analysis. The best-fit oscillation parameters are $|\Delta m^2_{23}|$ = 2.65×10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.98. The dotted line shows the NEUT prediction without oscillations.

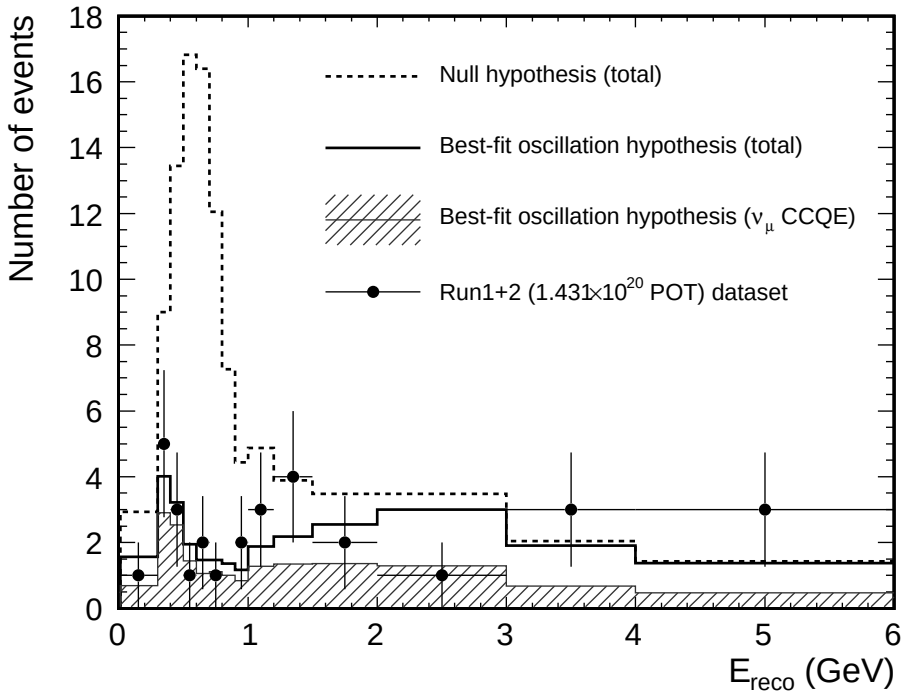


Figure 19: Same distributions as in Fig.17 with the addtion of the ν_μ-CCQE component of the best-fit oscillation hypothesis (GENIE)

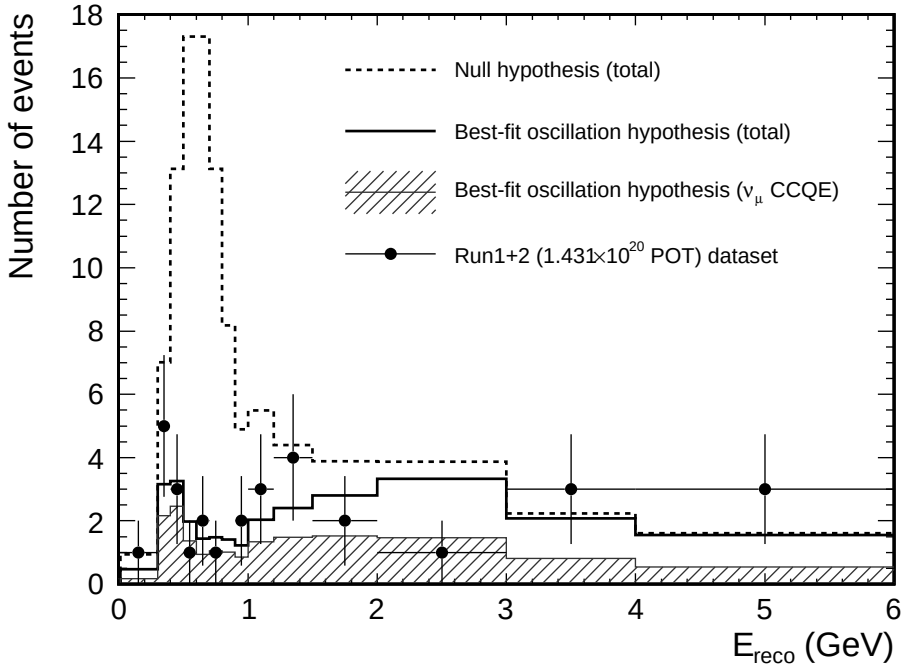


Figure 20: Same distributions as in Fig.18 with the addition of the ν_μ-CCQE component of the best-fit oscillation hypothesis (NEUT)

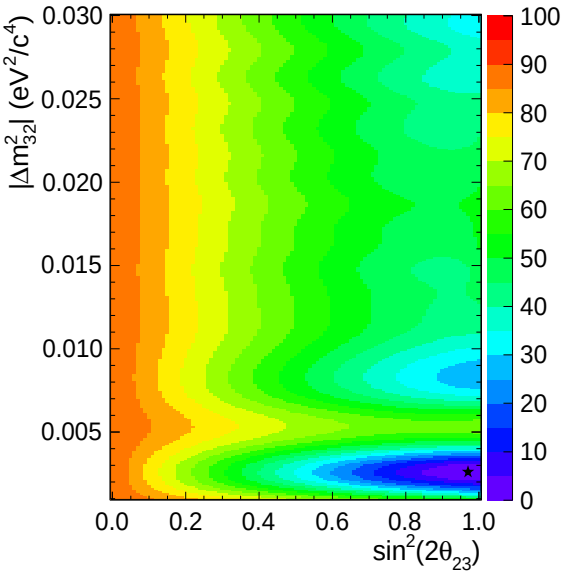


Figure 21: $\Delta\chi^2$ surface for the 2-flavour disappearance fit to the combined 1.431×10^{20} POT Run1+2 dataset (GENIE)

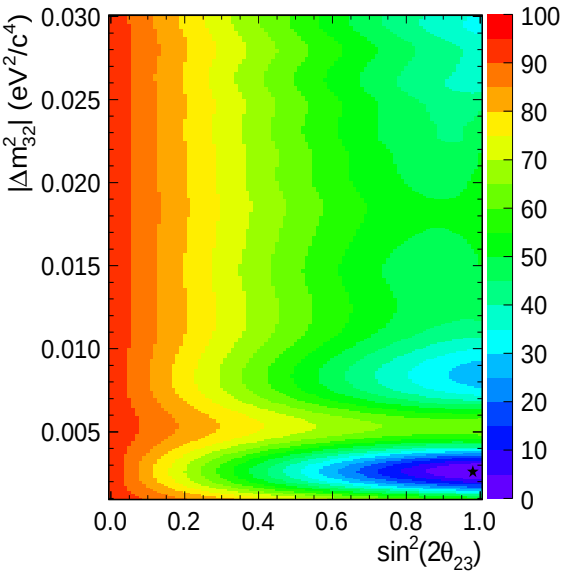


Figure 22: $\Delta\chi^2$ surface for the 2-flavour disappearance fit to the combined 1.431×10^{20} POT Run1+2 dataset (NEUT)

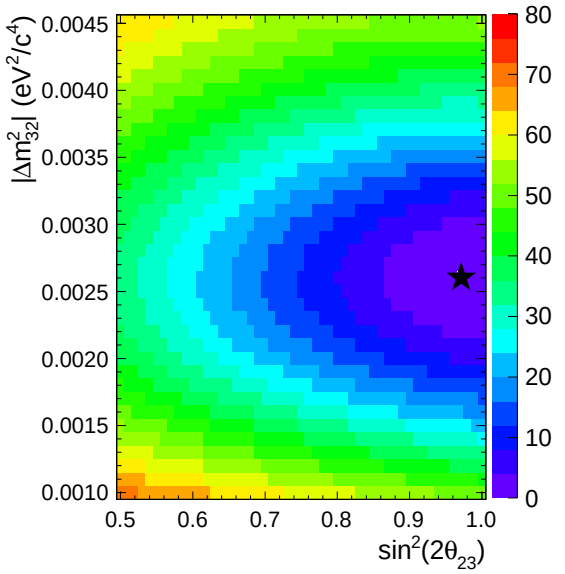


Figure 23: $\Delta\chi^2$ surface for the 2-flavour disappearance fit to the combined 1.431×10^{20} POT Run1+2 dataset, zoomed around the best-fit value (GENIE)

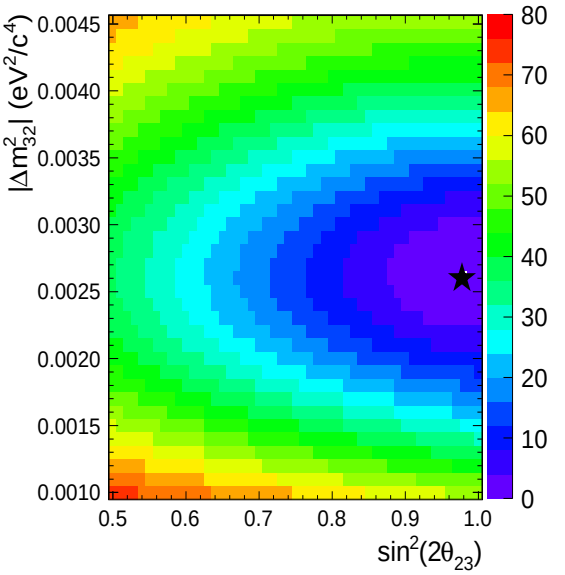


Figure 24: $\Delta\chi^2$ surface for the 2-flavour disappearance fit to the combined 1.431×10^{20} POT Run1+2 dataset, zoomed around the best-fit value (NEUT)

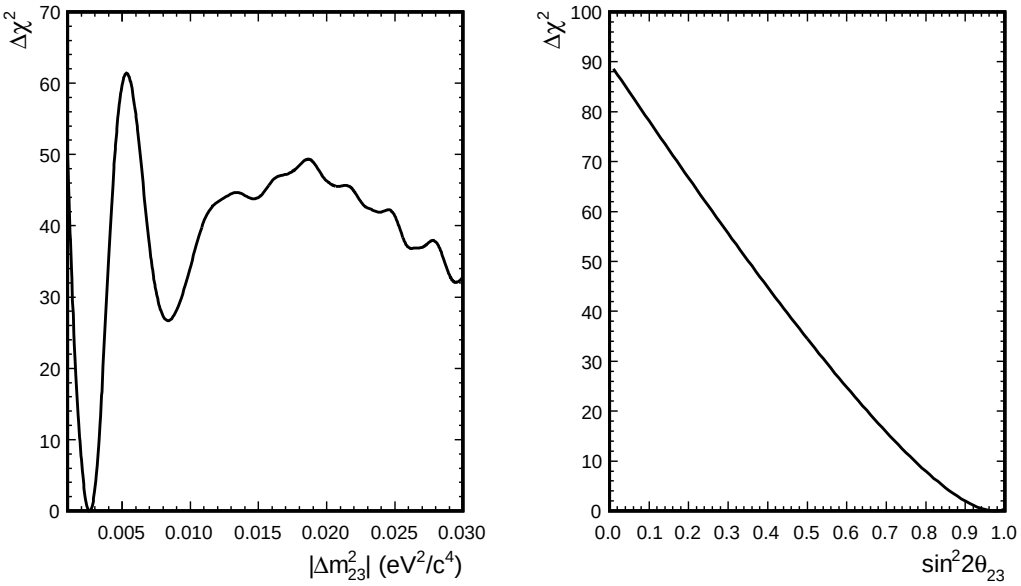


Figure 25: Marginalised $\Delta\chi^2$ distributions from the 2-flavour ν_μ-disappearance fit to the combined 1.431×10^{20} POT Run1+2 dataset (GENIE)

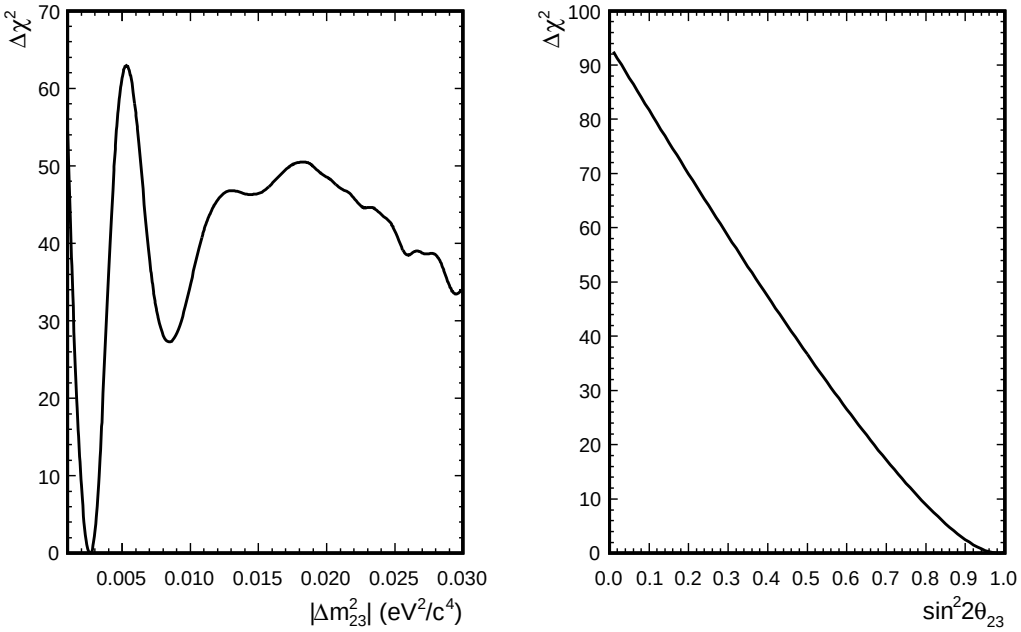


Figure 26: Marginalised $\Delta\chi^2$ distributions from the 2-flavour ν_μ-disappearance fit to the combined 1.431×10^{20} POT Run1+2 dataset (NEUT)

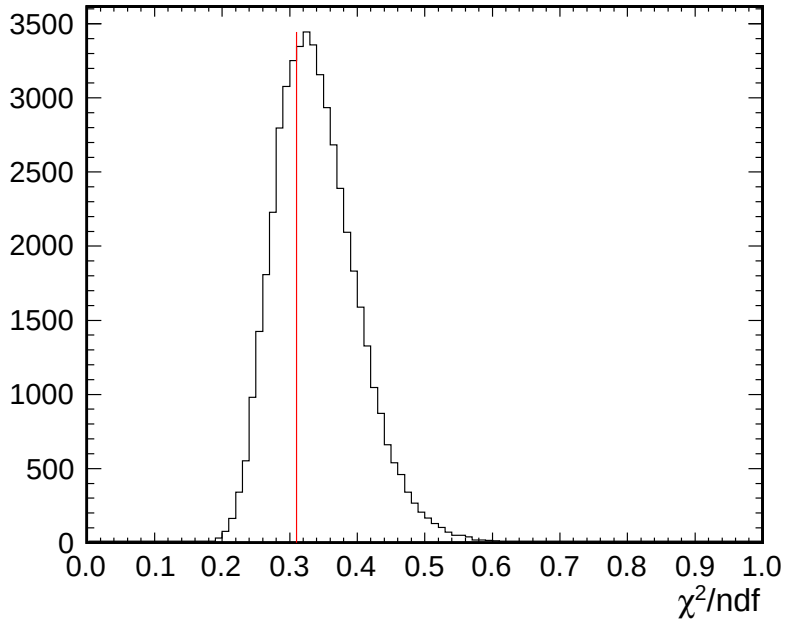


Figure 27: chi^2/ndf distribution for fits to 1.431×10^{20} POT toy MC experiments generated with $sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32×10^{-3} eV^2/c^4. The chi^2/ndf value for the fit to the Run1+2 dataset is also shown with a red line (GENIE).

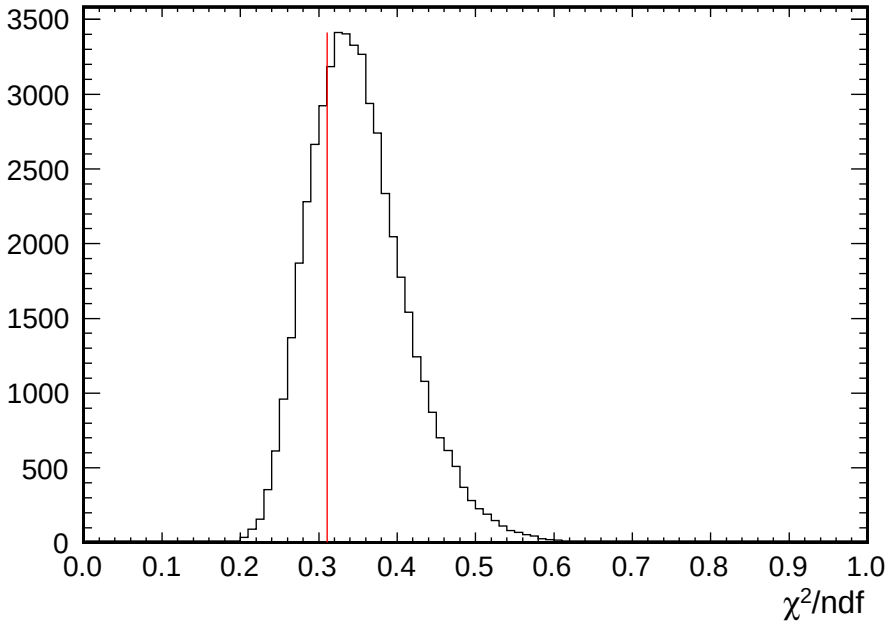


Figure 28: chi^2/ndf distribution for fits to 1.431×10^{20} POT toy MC experiments generated with $sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32×10^{-3} eV^2/c^4. The chi^2/ndf value for the fit to the Run1+2 dataset is also shown with a red line (NEUT).

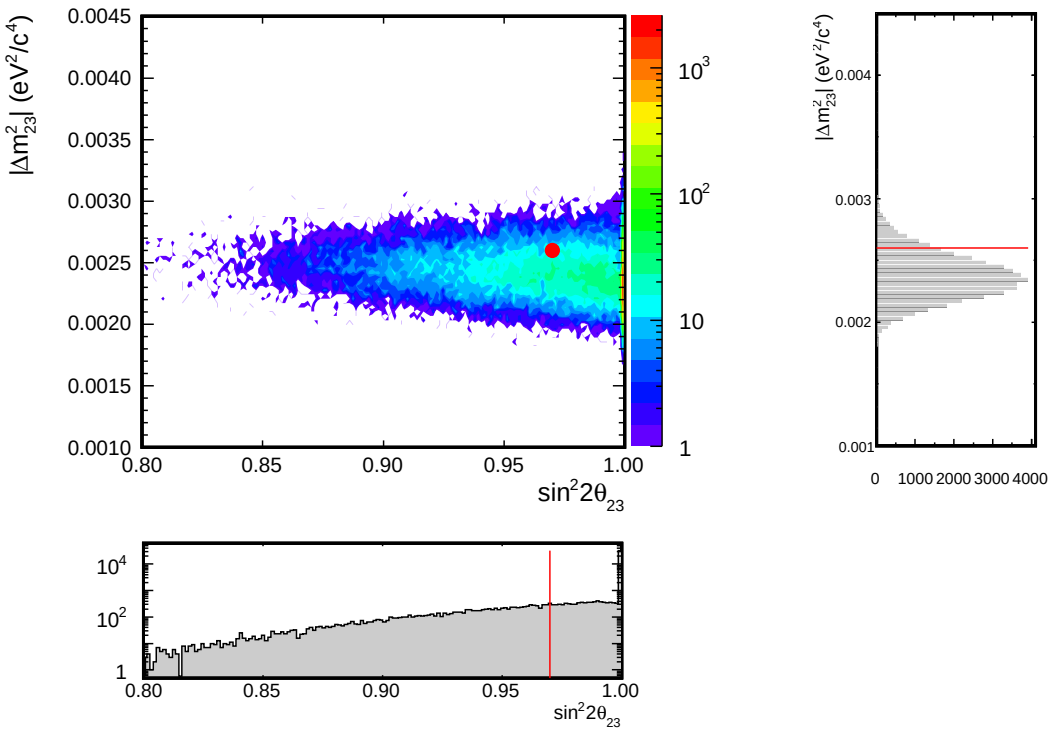


Figure 29: Best-fit parameter distributions from 2-flavour ν_μ-disappearance fits to 1.431×10^{20} POT toy MC experiments generated with $sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32×10^{-3} eV^2/c^4. The best-fit values for the fit to the Run1+2 dataset are also shown with a red marker and red lines (GENIE).

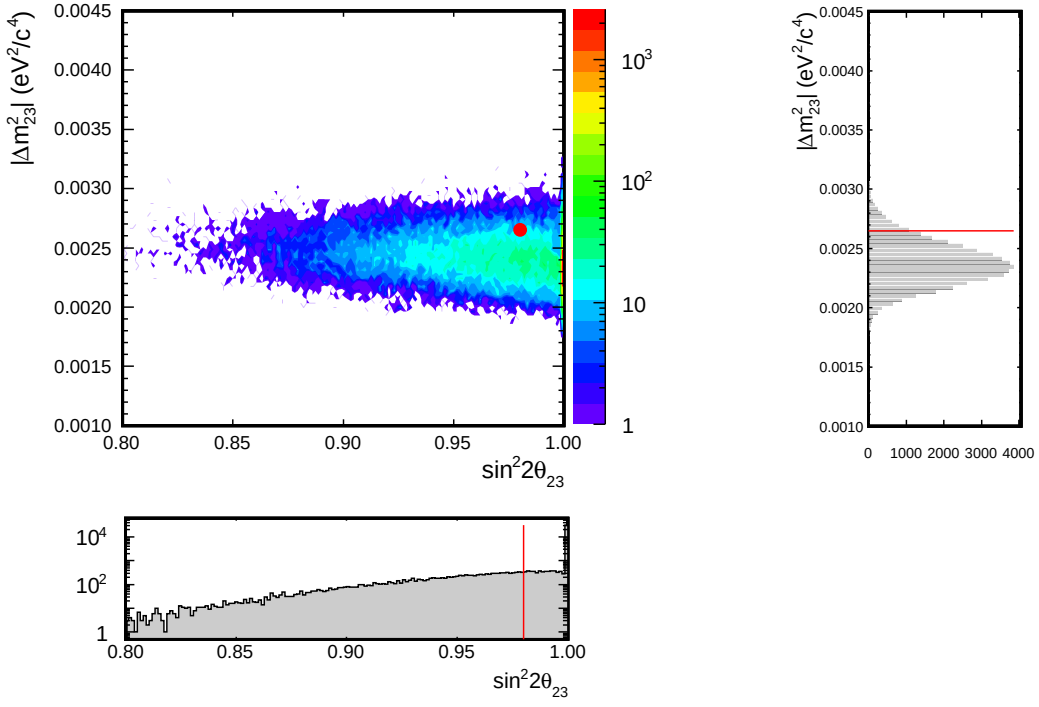


Figure 30: Best-fit parameter distributions from 2-flavour ν_μ-disappearance fits to 1.431×10^{20} POT toy MC experiments generated with $sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32×10^{-3} eV^2/c^4. The best-fit values for the fit to the Run1+2 dataset are also shown with a red marker and red lines (NEUT).

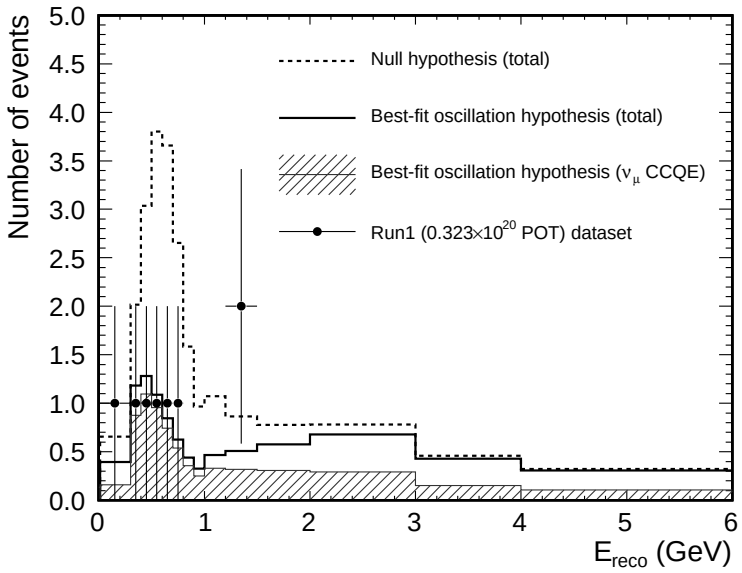


Figure 31: Reconstructed neutrino energy distribution of the 8 single μ-like ring events of the 0.323×10^{20} POT Run1 dataset (points) and GENIE best-fit SuperK prediction (solid line). The best-fit oscillation parameters are $|\Delta m^2_{23}|$ = 2.77×10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.81 (local minimum in first oscillation maximum region). The dotted line shows the GENIE prediction without oscillations. The distributions are not shown with 50 MeV bins as used in the oscillation fit, but with fewer, variable-size bins for illustration purposes.

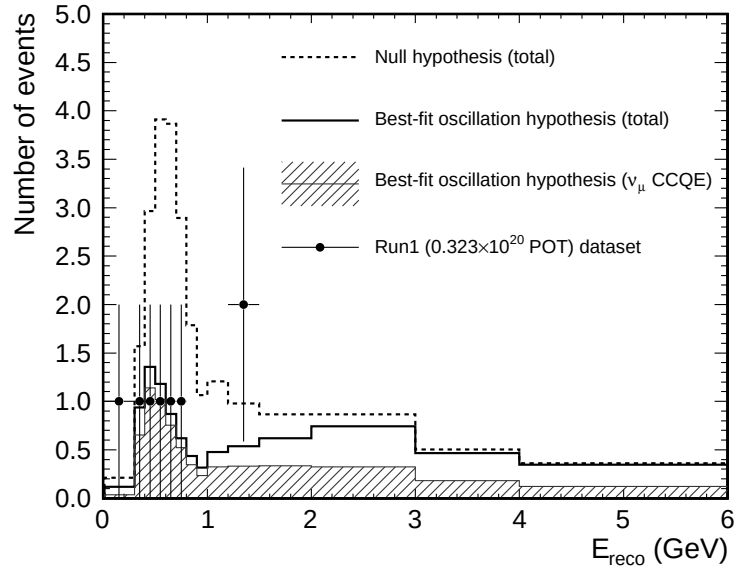


Figure 32: Reconstructed neutrino energy distribution of the 8 single μ-like ring events of the 0.323×10^{20} POT Run1 dataset (points) and NEUT best-fit SuperK prediction (solid line). The best-fit oscillation parameters are $|\Delta m^2_{23}|$ = 2.88×10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 0.84 (local minimum in first oscillation maximum region). The dotted line shows the NEUT prediction without oscillations. The distributions are not shown with 50 MeV bins as used in the oscillation fit, but with fewer, variable-size bins for illustration purposes.

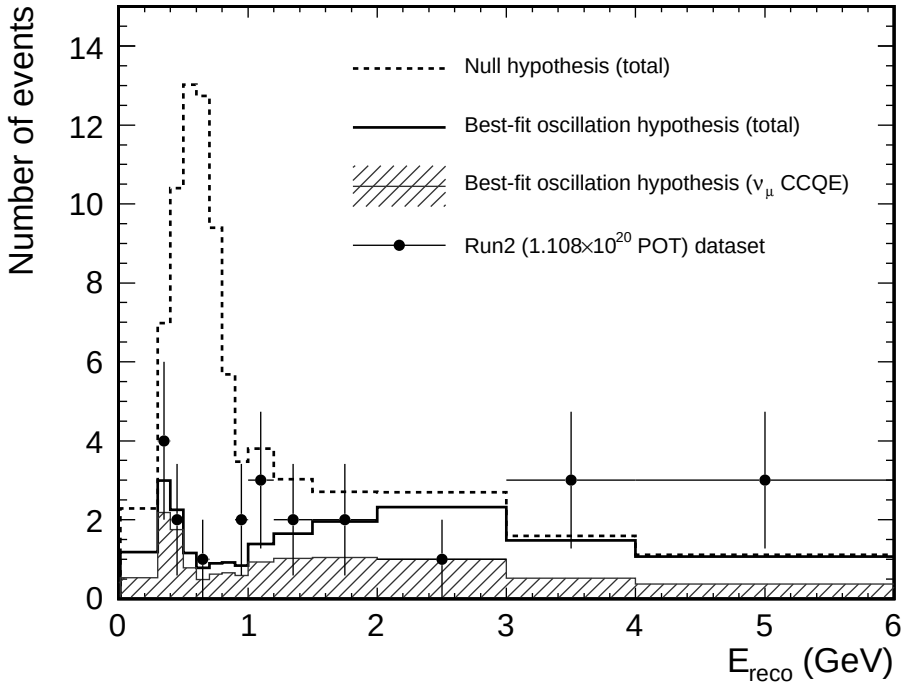


Figure 33: Reconstructed neutrino energy distribution of the 23 single μ-like ring events of the 1.108×10^{20} POT Run2 dataset (points) and GENIE best-fit SuperK prediction (solid line). The best-fit oscillation parameters are $|\Delta m^2_{23}|$ = 2.59×10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 1.00. The dotted line shows the GENIE prediction without oscillations. The distributions are not shown with 50 MeV bins as used in the oscillation fit, but with fewer, variable-size bins for illustration purposes.

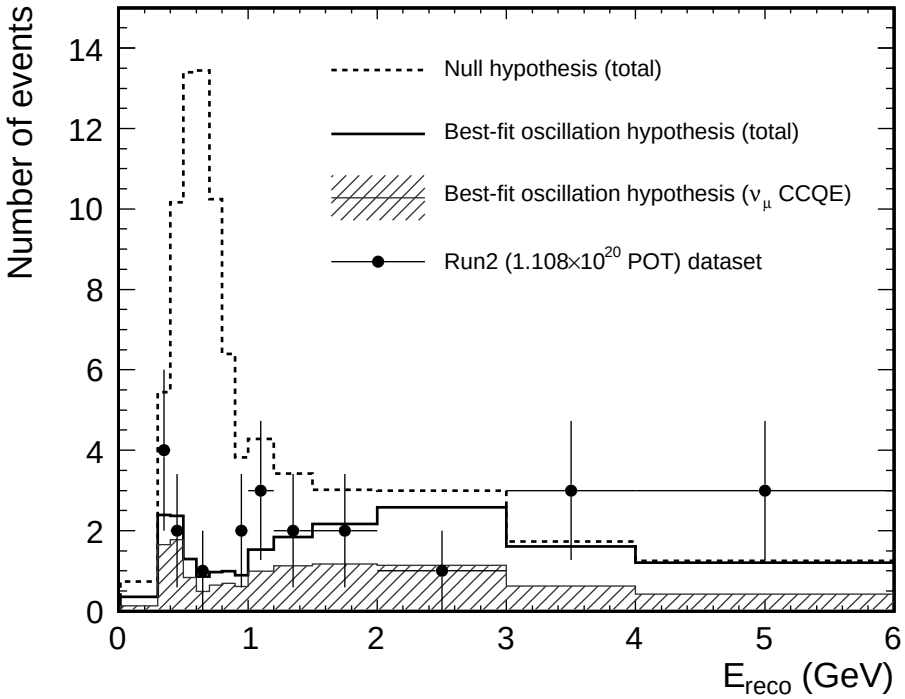


Figure 34: Reconstructed neutrino energy distribution of the 23 single μ-like ring events of the 1.108×10^{20} POT Run2 dataset (points) and NEUT best-fit SuperK prediction (solid line). The best-fit oscillation parameters are $|\Delta m^2_{23}|$ = 2.62×10^{-3} eV^2/c^4 and $sin^2 2\theta_{23}$ = 1.00. The dotted line shows the NEUT prediction without oscillations. The distributions are not shown with 50 MeV bins as used in the oscillation fit, but with fewer, variable-size bins for illustration purposes.

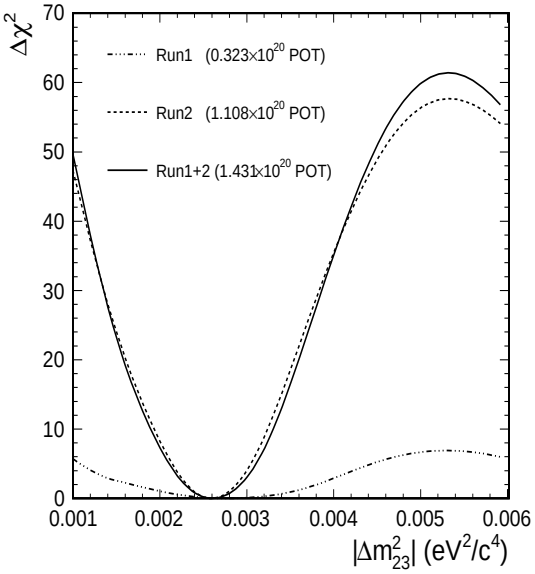


Figure 35: The marginalised $\Delta\chi^2$ distributions as a function of $|\Delta m^2_{23}|$ from the 2-flavour ν_μ-disappearance fits to the Run1, Run2 and Run1+2 datasets (GENIE). All fits were performed in the restricted $|\Delta m^2_{23}|$ range from $1\times10^{-3}\ eV^2/c^4$ to $6\times10^{-3}\ eV^2/c^4$.

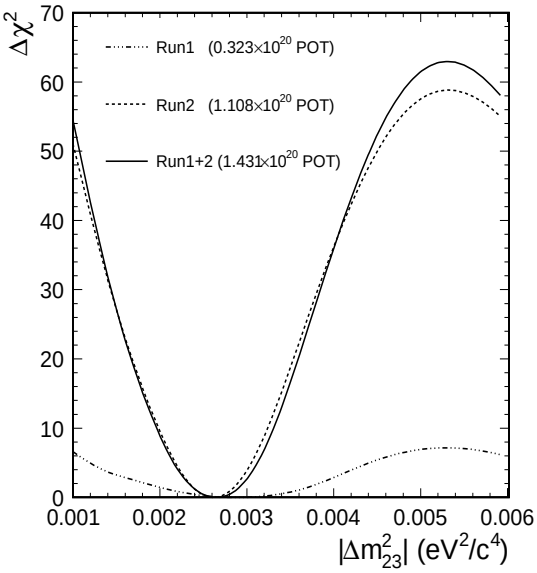


Figure 36: The marginalised $\Delta\chi^2$ distributions as a function of $|\Delta m^2_{23}|$ from the 2-flavour ν_μ-disappearance fits to the Run1, Run2 and Run1+2 datasets (NEUT). All fits were performed in the restricted $|\Delta m^2_{23}|$ range from $1\times10^{-3}\ eV^2/c^4$ to $6\times10^{-3}\ eV^2/c^4$.

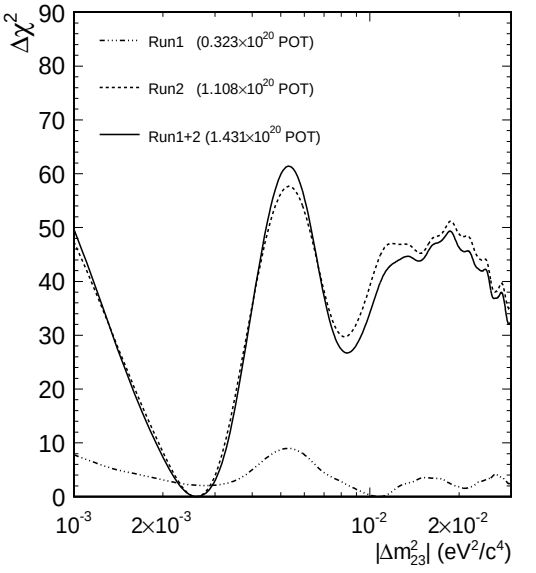


Figure 37: The marginalised $\Delta\chi^2$ distributions as a function of $|\Delta m^2_{23}|$ from the 2-flavour ν_μ-disappearance fits to the Run1, Run2 and Run1+2 datasets (GENIE). All fits were performed in the extended $|\Delta m^2_{23}|$ range from $1\times10^{-3}\ eV^2/c^4$ to $30\times10^{-3}\ eV^2/c^4$.

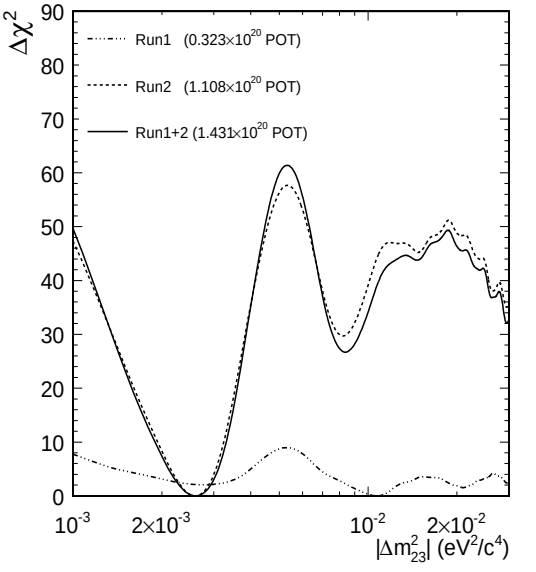


Figure 38: The marginalised $\Delta\chi^2$ distributions as a function of $|\Delta m^2_{23}|$ from the 2-flavour ν_μ-disappearance fits to the Run1, Run2 and Run1+2 datasets (NEUT). All fits were performed in the extended $|\Delta m^2_{23}|$ range from $1\times10^{-3}\ eV^2/c^4$ to $30\times10^{-3}\ eV^2/c^4$.

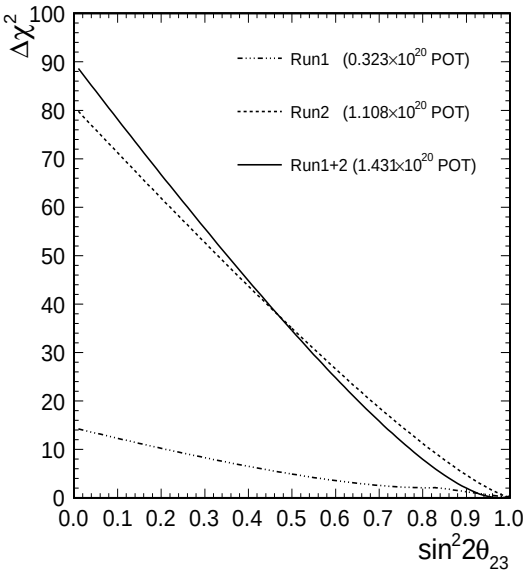


Figure 39: The marginalised $\Delta\chi^2$ distributions as a function of $sin^2 2\theta_{23}$ from the 2-flavour ν_μ-disappearance fits to the Run1, Run2 and Run1+2 datasets (GENIE). All fits were performed in the restricted $|\Delta m^2_{23}|$ range from 1×10^{-3} eV^2/c^4 to 6×10^{-3} eV^2/c^4.

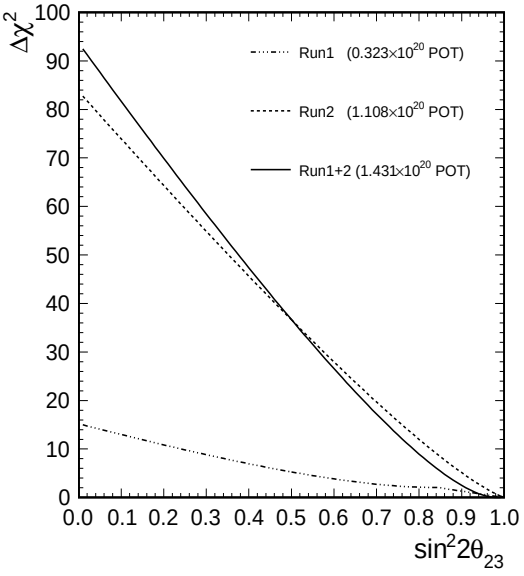


Figure 40: The marginalised $\Delta\chi^2$ distributions as a function of $sin^2 2\theta_{23}$ from the 2-flavour ν_μ-disappearance fits to the Run1, Run2 and Run1+2 datasets (NEUT). All fits were performed in the restricted $|\Delta m^2_{23}|$ range from 1×10^{-3} eV^2/c^4 to 6×10^{-3} eV^2/c^4.

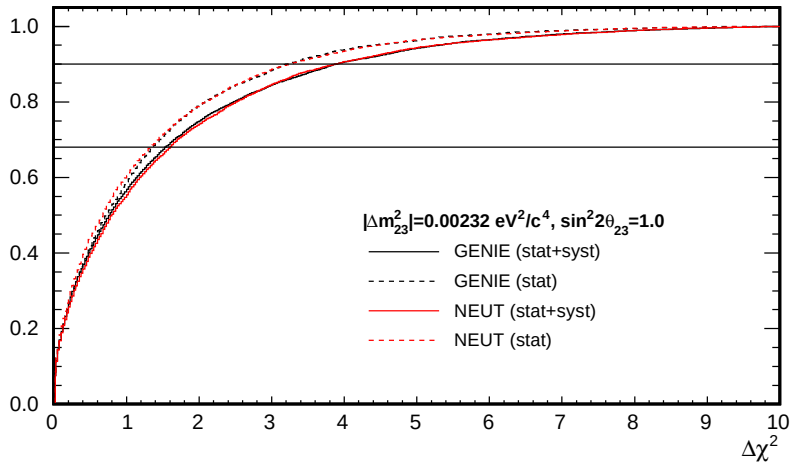


Figure 41: Cumulative $\Delta\chi^2_{MC}$ distributions (for both NEUT and GENIE) for oscillated toy MC experiments generated with $sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32×10^{-3} eV^2/c^4. The horizontal lines show the 68% and 90% levels (values of $\Delta\chi^2_{critical}$ are the values of $\Delta\chi^2_{MC}$ where these horizontal lines intersect with the cumulative distributions).

5. Construction of Feldman-Cousins confidence regions

The best-fit $|\Delta m^2_{23}|$ fluctuation seen in the analysis of the Run1 dataset is no longer present (see Figs. 37 and 38). The analysis of the Run2 and the combined Run1+2 datasets strongly disfavours any interpretation of the observed 1 μ-like ring event deficit in terms of large $|\Delta m^2_{23}|$ (a second or higher-order oscillation maximum at $\sim$600 MeV). Therefore, for reasons of *computational efficiency*, we limit the Feldman-Cousins construction to the $|\Delta m^2_{23}|$ range from 1×10^{-3} eV^2/c^4 to 6×10^{-3} eV^2/c^4.

The application of the Feldman-Cousins method [8] to the T2K ν_μ-disappearance analysis is discussed in detail in Sec. 3.4 of Ref. [4]: a number of toy MC experiment fits is performed for each point of the ($\sin^2 2\theta_{23}$, $|\Delta m^2_{23}|$) 2-dimensional grid. For each grid point, the critical value of $\Delta\chi^2$ ($\Delta\chi^2_{critical}$) is found such that α of the toy MC experiments have $\Delta\chi^2_{MC}$ $< \Delta\chi^2_{critical}$, where α is the desired confidence level (see Fig. 41). The value of $\Delta\chi^2_{MC}$ for each toy MC experiment is calculated as follows:

$$\Delta\chi^2_{MC} = \chi^2(\text{true}) - \chi^2(\text{MC, best-fit}) \tag{1}$$

where "true" means the true values of the oscillation parameters at that grid point, and "best-fit" means the best-fit values for that MC experiment (this is equivalent to equation 5.6 in [8]). χ^2 is defined as

$$\chi^2(sin^2 2\theta_{23}, |\Delta m^2_{23}|; \mathbf{a}) = 2 \cdot \sum_{i=0}^{N-1} \left(n_i^{obs} \cdot ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp} - n_i^{obs}) \right) \tag{2}$$

where N is the number of reconstructed energy bins, $\mathbf{a}$ is a vector of nuisance parameters (see Tab. 1), n_i^{obs} is the observed number of events in the ith bin, and $n_i^{exp} = n_i^{exp}(sin^2 2\theta_{23}, \Delta m^2_{23}; \mathbf{a})$ is the corresponding expected number of events. This definition of χ^2 arises from our use of the likelihood ratio method, which is explained in more detail in Sec. 3.2 of Ref. [4].

6. Run 1+2 oscillation parameter confidence regions

Feldman-Cousins confidence regions were constructed using the $\Delta\chi^2$($\sin^2 2\theta_{23}$, $|\Delta m^2_{23}|$) surfaces from the fits presented in Sec. 4 and the $\Delta\chi^2_{critical}$($\sin^2 2\theta_{23}$, $|\Delta m^2_{23}|$) maps presented in Sec. 5.

The 68% and 90% Feldman-Cousins confidence regions are shown for the 2-flavour ν_μ-disappearance fit to the combined Run1+2 dataset with and without systematics in Fig. 45 for GENIE and in Fig. 46 for NEUT. (It is worth noting that all confidence regions presented here are drawn using a $sin^2 2\theta_{23}$ bin size of 0.01 and a $|\Delta m^2_{23}|$ bin size of 1.0×10^{-4} eV^2/c^4). The star-shaped marker in each plot marks the bin with the minimum χ^2.) The 90% confidence regions are compared with those from MINOS [3] and SuperK [18] in Fig. 47 for GENIE and in Fig. 48 for NEUT. A similar comparison is shown in Figs. 49 and 50, and includes the 68% confidence regions from T2K and MINOS.

In Fig. 51 the 90% Feldman-Cousins confidence region (with systematics) from the GENIE 2-flavour ν_μ-disappearance fit to the combined Run1+2 dataset is compared with the T2K 1.431×10^{20} POT sensitivity for oscillations with the 2011 MINOS best-fit values ($sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}|$ = 2.32×10^{-3} eV^2/c^4). The 2011 MINOS 90% confidence region is also shown in this figure. The equivalent comparison from the NEUT 2-flavour ν_μ-disappearance fit is shown in Fig. 52.

The 90% Feldman-Cousins confidence region from the fit to the combined Run1+2 dataset is compared with the 90% confidence regions from the fits to the Run1-only and Run2-only datasets in Fig. 53 for GENIE and in Fig. 54 for NEUT.

Finally, a comparison of the GENIE and NEUT 90% Feldman-Cousins confidence regions from the fit to the combined Run1+2 dataset is shown in Fig. 55.

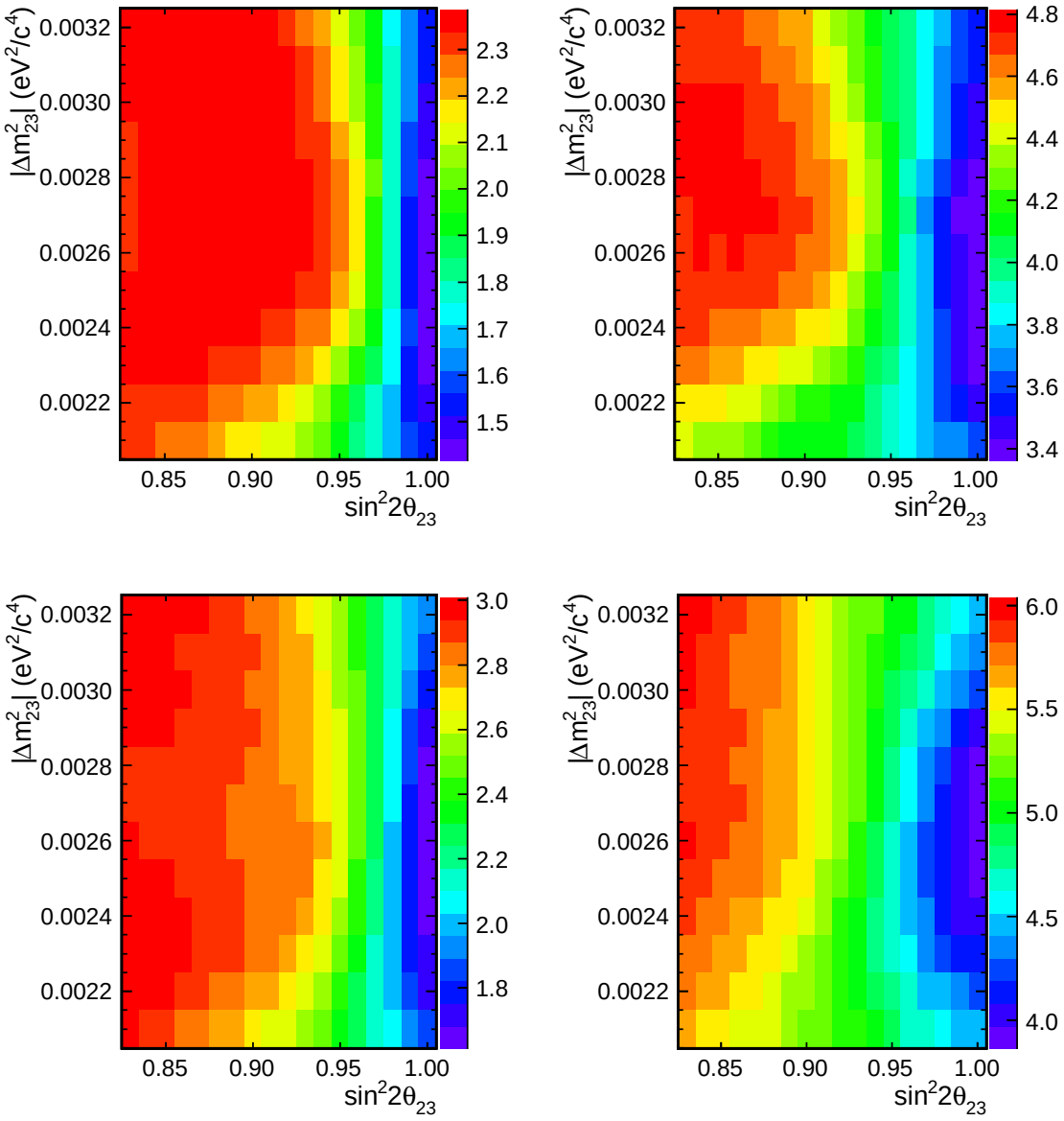


Figure 42: Monte Carlo estimation, according to the Feldman-Cousins method, of $\Delta\chi^2_{68}$ (left) and $\Delta\chi^2_{90}$ (right) as functions of $sin^2 2\theta_{23}$ and Δm^2_{23}, for calculation of the 68% (left) and 90% (right) CL confidence regions for the 1.431×10^{20}-POT T2K 2-flavour ν_μ-disappearance analysis. The top row shows $\Delta\chi^2$ surfaces calculated using statistical fluctuations only whereas the bottom row shows $\Delta\chi^2$ surfaces calculated using both systematic variations and statistical fluctations. As expected from [8], the values of $\Delta\chi^2$ decrease as the physical boundary at $sin^2 2\theta_{23} = 1$ is approached.

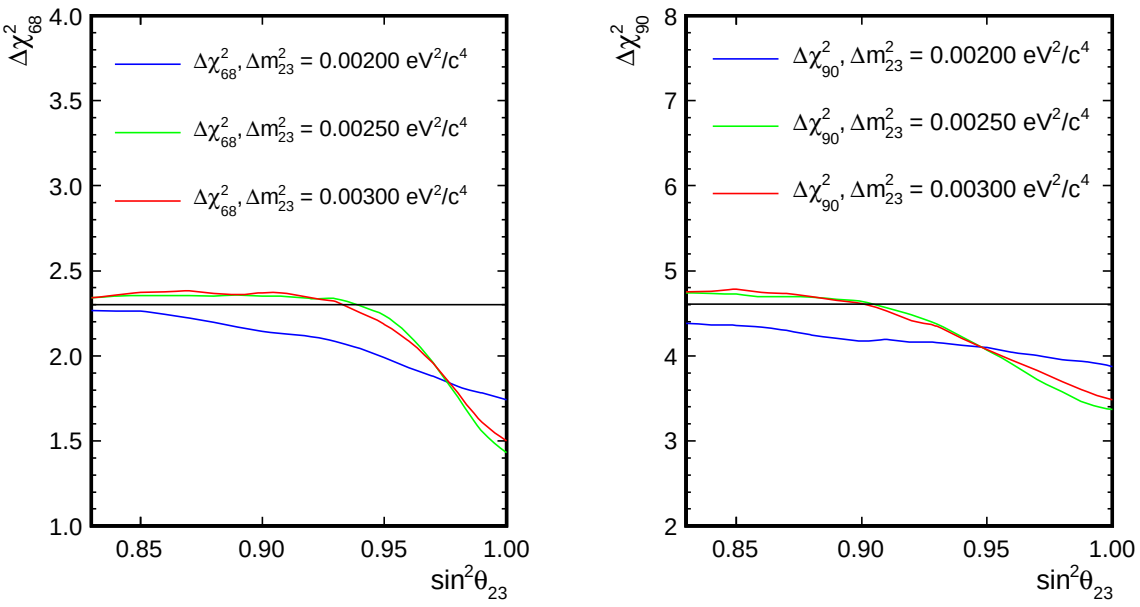


Figure 43: Monte Carlo estimation, according to the Feldman-Cousins method (statistical errors only), of $\Delta\chi^2_{68}$ (left) and $\Delta\chi^2_{90}$ (right), as functions of $sin^2 2\theta_{23}$ for 3 different values of Δm^2_{23} ($2.0 \times 10^{-3} eV^2/c^4$, $2.5 \times 10^{-3} eV^2/c^4$ and $3.0 \times 10^{-3} eV^2/c^4$). For comparison, the canonical value of $\Delta\chi^2$ used in the constant $\Delta\chi^2_{critical}$ method is also shown as a horizontal line.

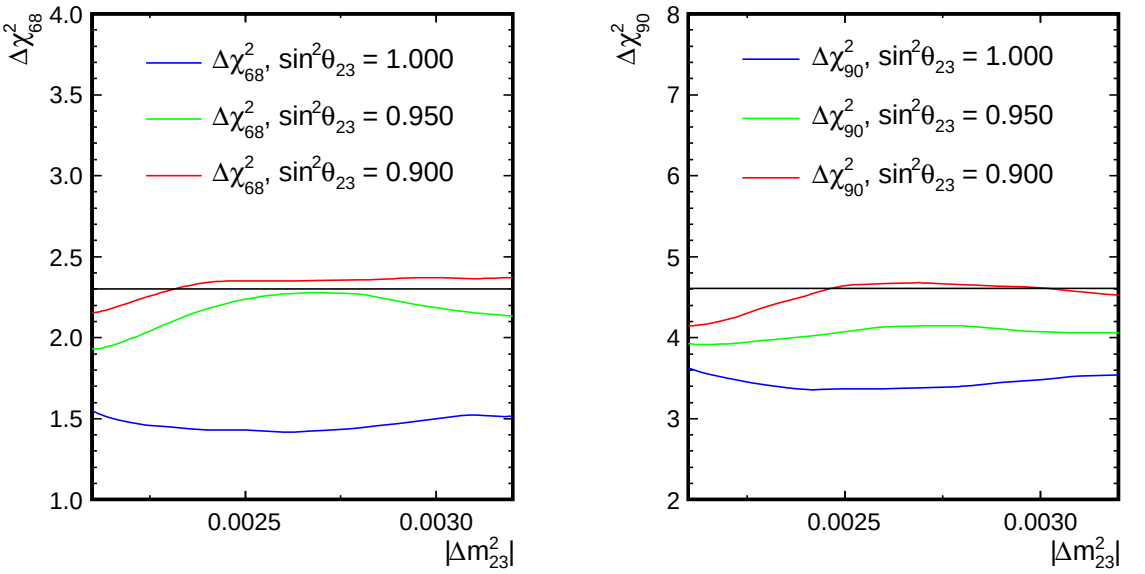


Figure 44: Monte Carlo estimation, according to the Feldman-Cousins method (statistical errors only), of $\Delta\chi^2_{68}$ (left) and $\Delta\chi^2_{90}$ (right), as functions of Δm^2_{23} for 3 different values of $sin^2 2\theta_{23}$ (0.90, 0.95 and 1.00). For comparison, the canonical value of $\Delta\chi^2$ used in the constant $\Delta\chi^2_{critical}$ method is also shown as a horizontal line.

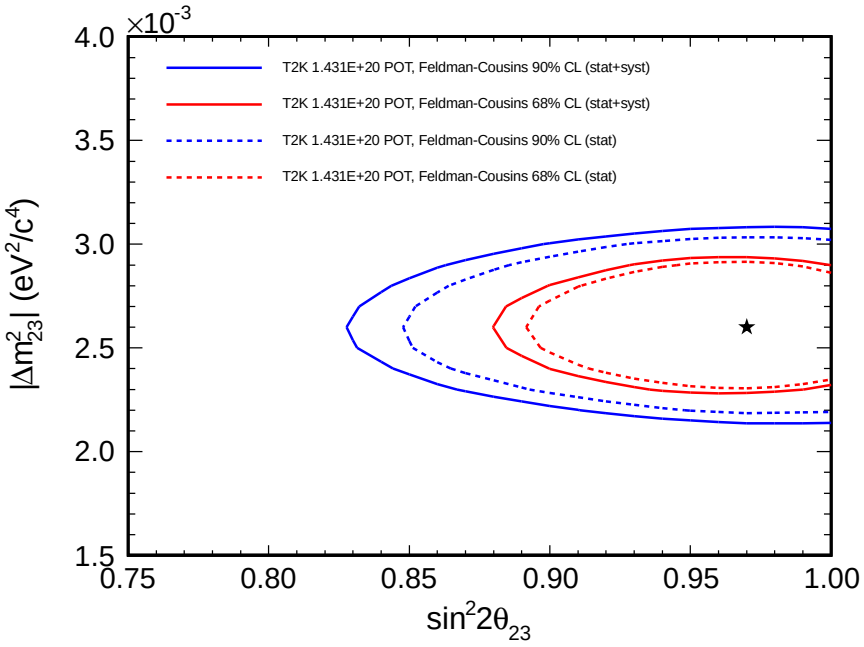


Figure 45: 68% and 90% Feldman-Cousins confidence regions, with and without systematics, for the GENIE 2-flavour ν_μ-disappearance fit to the combined Run1+2 dataset

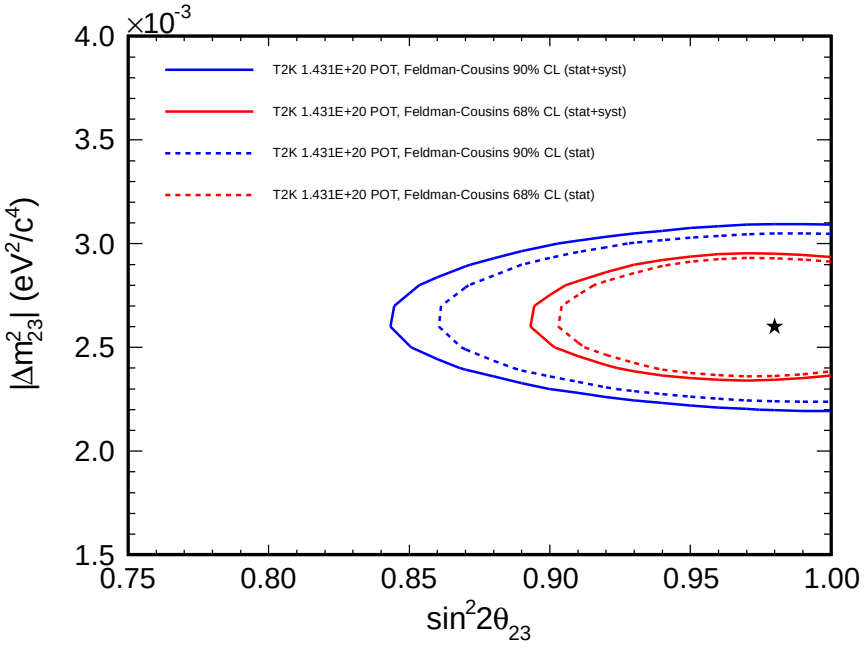


Figure 46: 68% and 90% Feldman-Cousins confidence regions, with and without systematics, for the NEUT 2-flavour ν_μ-disappearance fit to the combined Run1+2 dataset

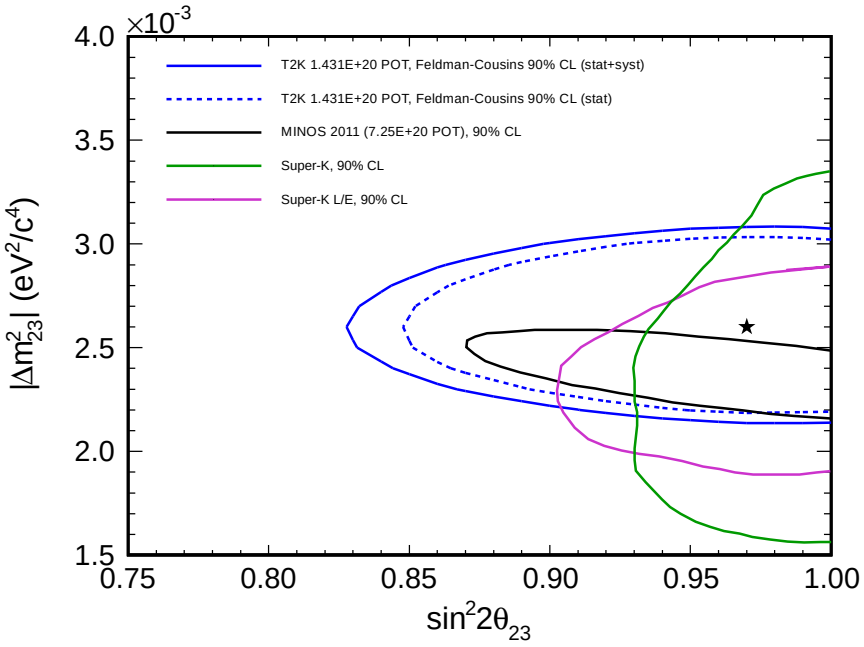


Figure 47: 90% Feldman-Cousins confidence regions, with and without systematics, for the GENIE 2-flavour ν_μ-disappearance fit to the combined Run1+2 dataset: comparisons with results from MINOS [3] and SuperK [18].

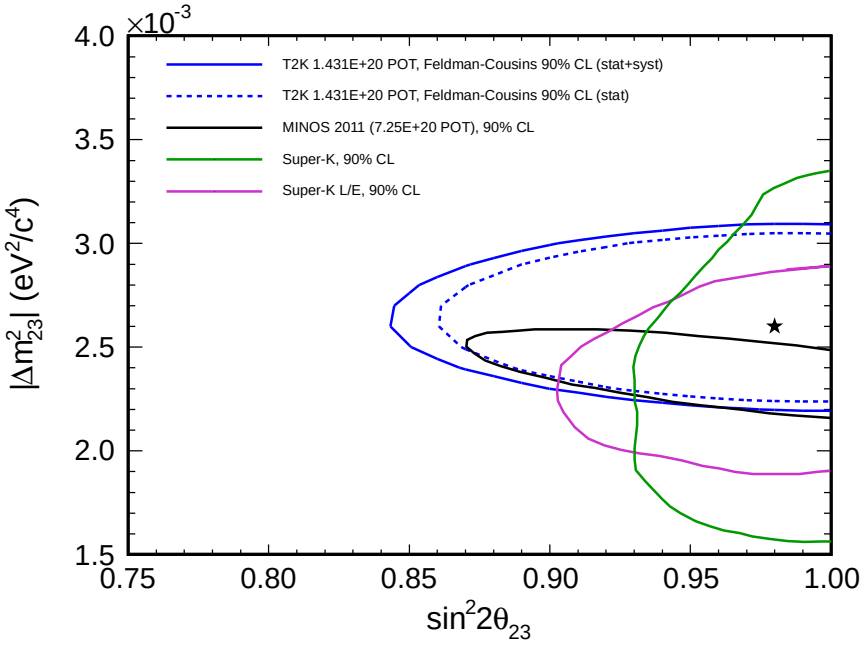


Figure 48: 90% Feldman-Cousins confidence regions, with and without systematics, for the NEUT 2-flavour ν_μ-disappearance fit to the combined Run1+2 dataset: comparisons with results from MINOS [3] and SuperK [18].

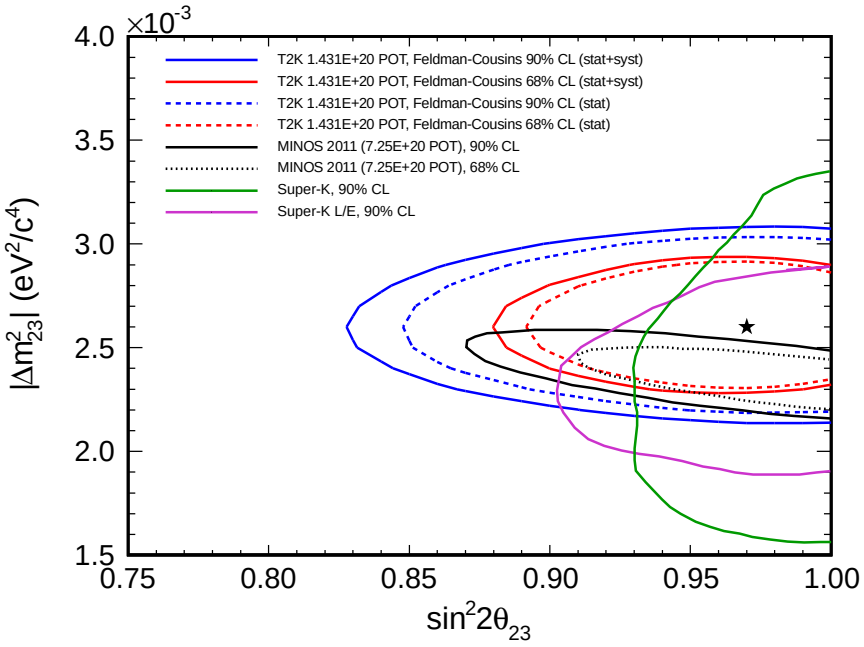


Figure 49: 68% and 90% Feldman-Cousins confidence regions, with and without systematics, for the GENIE 2-flavour ν_μ-disappearance fit to the combined Run1+2 dataset: comparisons with MINOS 2011 68% and 90% contours [3] and SuperK 90% contour [18].

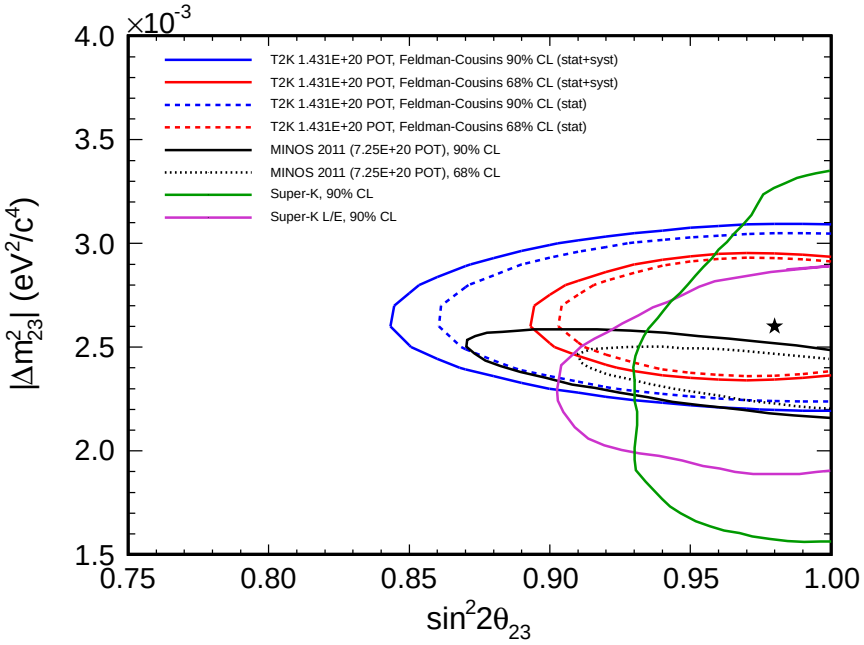


Figure 50: 68% and 90% Feldman-Cousins confidence regions, with and without systematics, for the NEUT 2-flavour ν_μ-disappearance fit to the combined Run1+2 dataset: comparisons with MINOS 2011 68% and 90% contours [3] and SuperK 90% contour [18].

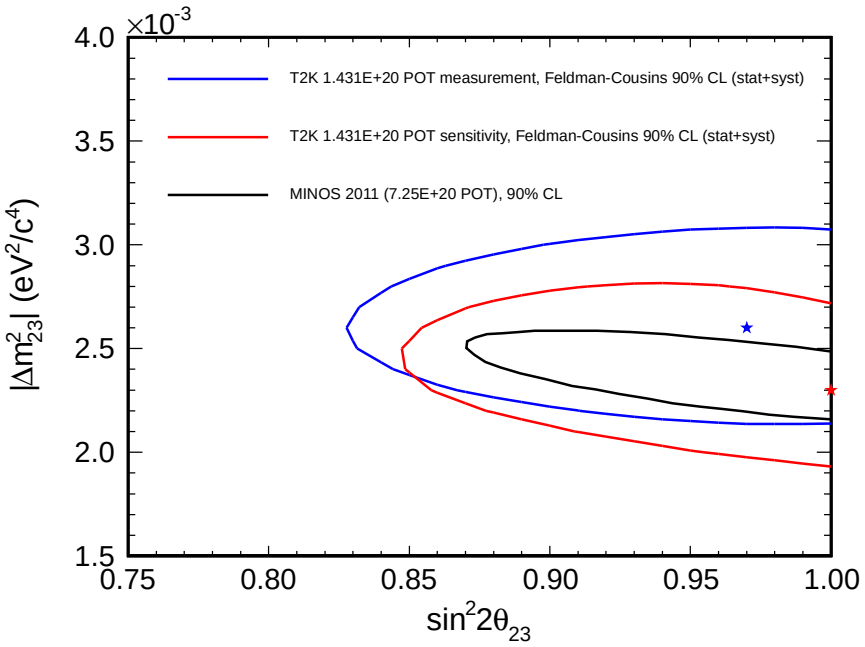


Figure 51: 90% Feldman-Cousins confidence regions (with systematics) for the GENIE 2-flavour ν_μ-disappearance fit to the combined Run1+2 dataset: comparisons with T2K 1.431×10^{20} POT sensitivity for ν_μ-disappearance with the MINOS 2011 best-fit values ($sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}| = 2.32\times10^{-3}$ eV^2/c^4). The MINOS 2011 90% confidence region [3] is also shown.

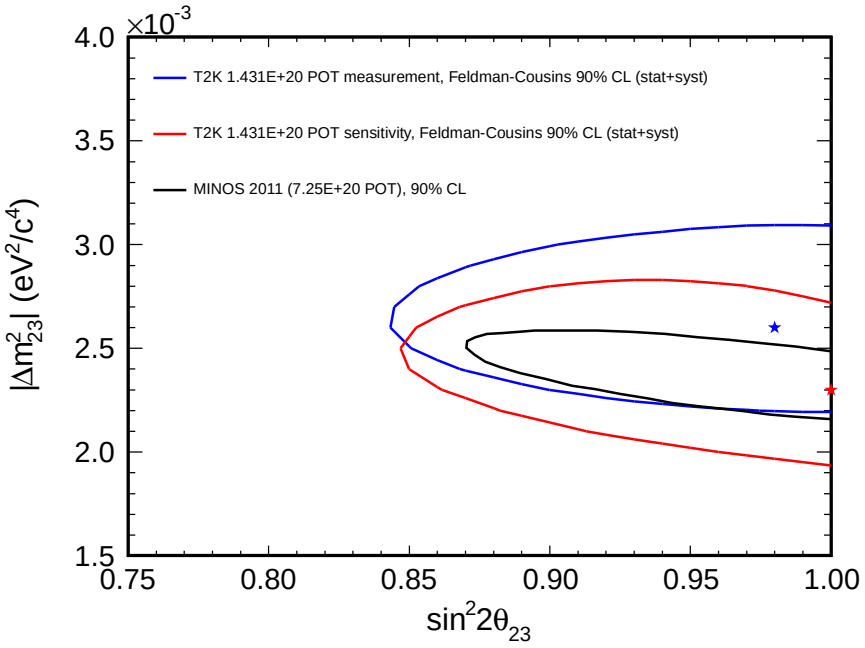


Figure 52: 90% Feldman-Cousins confidence regions (with systematics) for the NEUT 2-flavour ν_μ-disappearance fit to the combined Run1+2 dataset: comparisons with T2K 1.431×10^{20} POT sensitivity for ν_μ-disappearance with the MINOS 2011 best-fit values ($sin^2 2\theta_{23}$ = 1.0 and $|\Delta m^2_{23}| = 2.32\times10^{-3}$ eV^2/c^4). The MINOS 2011 90% confidence region [3] is also shown.

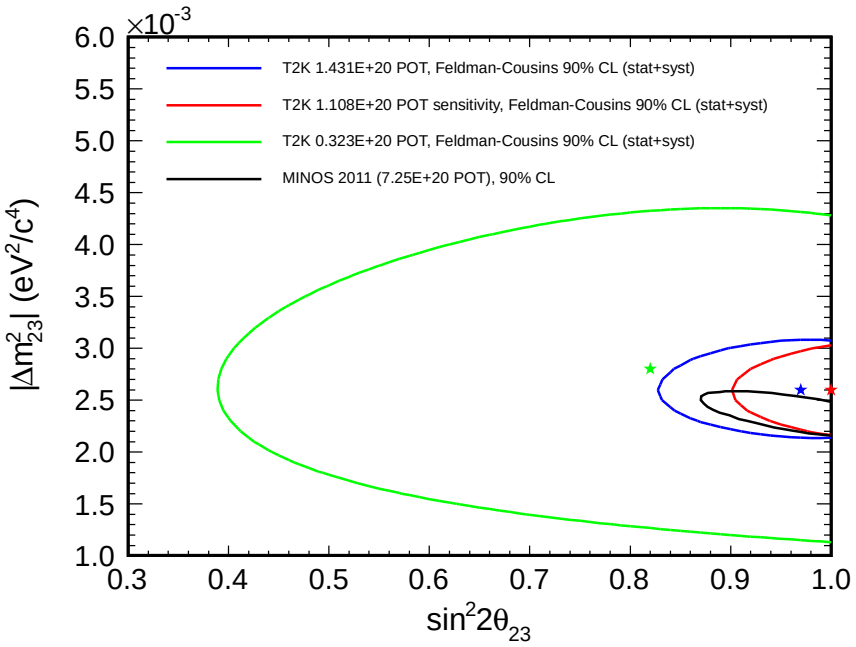


Figure 53: 90% Feldman-Cousins confidence regions (with systematics) for the GENIE 2-flavour ν_μ-disappearance fits to the Run1, Run2 and the combined Run1+2 datasets. The MINOS 2011 result [3] is also shown for reference.

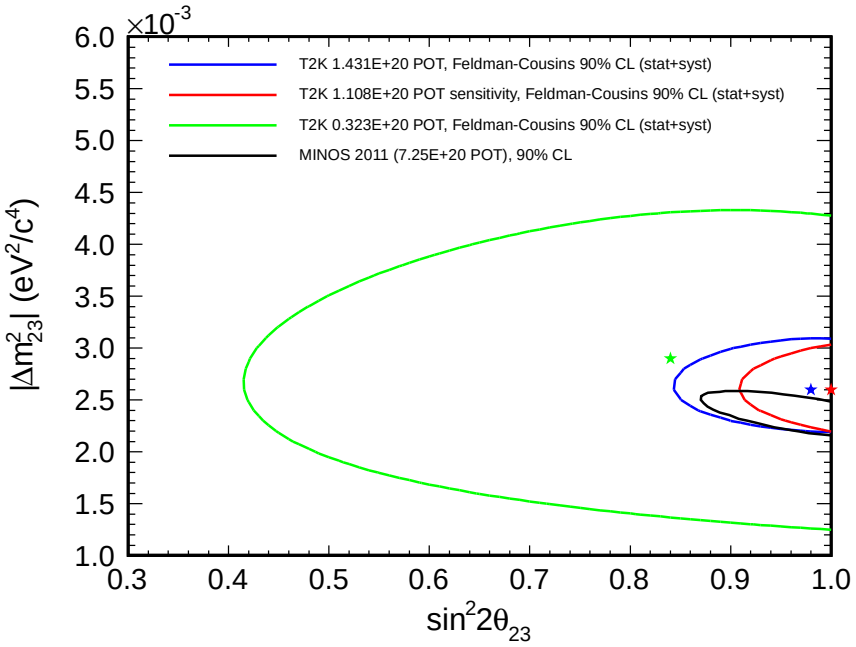


Figure 54: 90% Feldman-Cousins confidence regions (with systematics) for the NEUT 2-flavour ν_μ-disappearance fits to the Run1, Run2 and the combined Run1+2 datasets. The MINOS 2011 result [3] is also shown for reference.

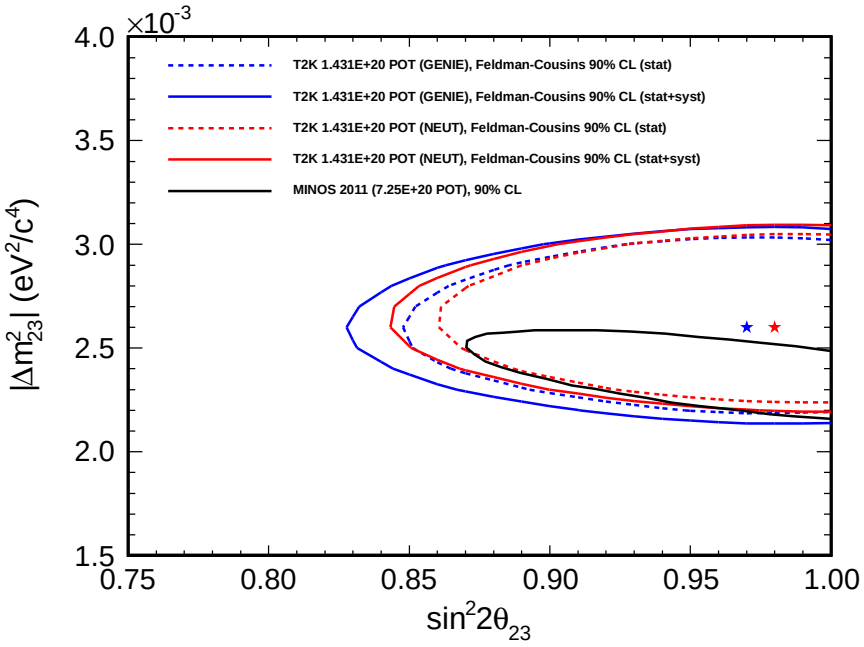


Figure 55: Comparison of the GENIE and NEUT 90% Feldman-Cousins confidence regions (with and witout systematics) for the 2-flavour ν_μ-disappearance fits to the combined Run1+2 dataset. The MINOS 2011 result [3] is also shown for reference.

7. Summary

In this note, we have presented the results of a 2-flavour ν_μ-disappearance analysis on the combined Run 1+2 dataset, which corresponds to an integrated JPARC neutrino beam exposure of 1.431×10^{20} POT. The analysis method was identical to that used in our earlier analysis reported in Ref. [4] except that updates were used of the SuperK flux, the flux normalisation error (N_{SK}/N_{ND}), the flux shape error, the ND280 event rate normalisation correction, the SuperK ring-counting efficiency error and the SuperK energy scale error.

The ND280 reconstructed energy spectrum was extrapolated to predict the number of single μ-like ring events that would be expected in SuperK if there were no oscillations. With GENIE this prediction was 101.72 ± 10.09 (stat) $^{+13.88}_{-13.49}$ (syst), while NEUT predicted 103.59 ± 10.18 (stat) $^{+13.77}_{-13.43}$ (syst) events. Only 31 single μ-like ring events were observed in SuperK, and the statistical significance of the deficit was $\sim$4.2 standard deviations for GENIE and $\sim$4.3 standard deviations for NEUT.

A 2-flavour ν_μ-disappearance fit to the observed SuperK reconstructed energy spectrum of single μ-like ring events in the combined Run 1+2 dataset gave best-fit values of $\sin^2(2\theta_{23}) = 0.97$ and $|\Delta m^2_{23}| = 2.60 \times 10^{-3}$ eV^2/c^4 (χ^2/ndf = 0.310) for GENIE and $\sin^2(2\theta_{23}) = 0.98$ and $|\Delta m^2_{23}| = 2.65 \times 10^{-3}$ eV^2/c^4 (χ^2/ndf = 0.311) for NEUT. Separate 2-flavour ν_μ-disappearance fits were also made to each of the Run 1 and Run 2 datasets. For Run 1 these gave best-fit values of $\sin^2(2\theta_{23}) = 0.81$ and $|\Delta m^2_{23}| = 2.77 \times 10^{-3}$ eV^2/c^4 (χ^2/ndf = 0.105) for GENIE and $\sin^2(2\theta_{23}) = 0.84$ and $|\Delta m^2_{23}| = 2.88 \times 10^{-3}$ eV^2/c^4 (χ^2/ndf = 0.114) for NEUT. Meanwhile the best-fit values for Run 2 were $\sin^2(2\theta_{23}) = 1.00$ and $|\Delta m^2_{23}| = 2.59 \times 10^{-3}$ eV^2/c^4 (χ^2/ndf = 0.318) for GENIE and $\sin^2(2\theta_{23}) = 1.00$ and $|\Delta m^2_{23}| = 2.62 \times 10^{-3}$ eV^2/c^4 (χ^2/ndf = 0.310) for NEUT.

Confidence contours for the oscillation parameters $\sin^2(2\theta_{23})$ and $|\Delta m^2_{23}|$ were constructed using the Feldman-Cousins method. These contours have been presented at 68% and 90% confidence both with and without systematic errors for the combined Run 1+2 dataset. Contours have also been shown at 90% confidence with systematic errors for each of Run 1 and Run 2 separately. The 68% and 90% confidence contours with and without systematic errors for the combined Run 1+2 dataset have been compared with previous results from MINOS and SuperK.

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